ROYAL GLOBAL UNIVERSITY
GUWAHATI

(ROYAL SCHOOL OF MEDICAL & ALLIED SCIENCES)

(RSMAS)

DEPARTMENT OF PHYSIOTHERAPY

Course structure & Syllabus

In Line with National Commission for Allied and Healthcare Professions (NCAHP)

For

MASTERS OF PHYSIOTHERAPY

W.E.F 2026 - 27

Curriculum

MASTER OF PHYSIOTHERAPY

[M.P.T]

1.18 SCHEME OF STUDY MASTER OF PHYSIOTHERAPY (M.P.T.)

1.18.1. First Year M.P.T Examination Scheme

Table 5.4. First Year M.P.T Examination Scheme

S. No.	Subject	Internal Assessment Marks		University Examination Marks			Total Marks	Theory hours	Practical hours	Total Hours	Credits Theory	Credits Practical	Credits Total
		Theory	Practical	Theory	Viva	Practical							
1	**PHT244C101** Laws, Ethics, Administratio n Educational methodology (LEM)			100			100	90		90	6		6
2	**PHT244C102** Research methodology and biostatistics, EBP (RMB)			100			100	90		90	6		6
3	**PHT244C103** Biomechanics & Therapeu-Tics (BCT)			100			100	90		90	6		6
4	**PHT244D10M1/PHT 244D10N1/PHT244 D10C1** *Physical & Functional Diagnosis in the speciality.* Speciality paper-1			100			100	120	120	240	8	4	12
5	**PHT244C111** Skills acquisition and refinement (SAR-I)	(Teaching Assignment, Seminars, journal club & Case Studies etc.)							240	240		8	8

S. No.	Subject	Internal Assessment Marks		University Examination Marks			Total Marks	Theory hours	Practical hours	Total Hours	Credits Theory	Credits Practical	Credits Total
		Theory	Practical	Theory	Viva	Practical							
6	**PHT244C112** Clinical training (CT-I)								690	690		23	23
7	**PHT244C121** Dissertation (DSS-I)								90	90		3	3
	Grand Total						400	390	1140	1530	26	38	64

i. N.B.-The [NUE] Subjects will on college level and students needs to pass the college level examination before appearing for the University Examination, But the marks will be counted with University Marks and will be added in the Scheme and Marks Sheet given by University.

SCHEME OF STUDY MASTER OF PHYSIOTHERAPY (M.P.T.)

1.18.2. 2ND Year M.P.T Examination

Table 5.5. Second Year M.P.T Examination Scheme

S. No.	Subject	Internal Assessment Marks		University Examination Marks			Total Marks	Theory hours	Practical hours	Total Hours	Credits Theory	Credits Practical	Credits Total
		Theory	Practical	Theory	Viva	Practical							
1	**PHT244C201** Exercise Physiology (EP)			100			100	90		90	6	0	6
2	**PHT244D20M2/PHT244D20N2/PHT244D20C2** *Specialty Paper 2*			100			100	120	120	240	8	4	12

3	**PHT244D20M3/PHT244D20N3/PHT244D20C3** *Specialty paper 3(Recent advances in the specialty)*			100			100	120	120	240	8	4	12
4	**PHT244C221** Dissertation[spread over a period of 18 months](DSS-II)								**660**	**660**			**22**
5	**PHT244C211** Skills acquisition and refinement(SAR-II)	(Teaching assignment, Seminar, journal club & Case Studies etc.)							**240**	**240**		**8**	**8**
6	**PHT244C212** Clinical training(CT-II)								**660**	**600**		**20**	**20**
	Theory Total						**300**	**330**	**1740**	**2070**	**22**	**58**	**80**
Practical Marks													
7	**PHT244D21M2/PHT244D21N2/PHT244D21C2** Major Elective					**150**	**150**						
8	**PHT244D21M3/PHT244D21N3/PHT244D21C3** Two Minor Elective					**150**	**150**						
9	**PHT244C213** OPSE/OSCE	**Clinical and Practical Exam**				**100**	**100**						

10	**PHT244C221** Dissertation		**50**	**50**						
	Practical Total		**450**	**450**						
Grand Total				**750**	**330**	**1740**	**2070**	**22**	**58**	**80**

N.B.-

i. Viva marks will be added in theory marks along with internal assessment theory; candidate have to get min. 50% marks in theory and viva collectively for passing the examination.

ii. The [NUE] Subjects will on college level and students needs to pass the college level examination before appearing for the University Examination, But the marks will be counted with University Marks and will be added in the Scheme and Marks Sheet given by university.

1.19. Curriculum Outline and detailed Curriculum

1. **Common subjects for all PG**
    1. Laws, Ethics, Administration and Educational methodology
    2. Research methodology, biostatistics and EBP
    3. Biomechanics and Therapeutics
    4. Locomotor Disability Assessment : *To be taught in First year as a part of syllabus*
    5. BLS and ALS- *To be taught in First year as a part of syllabus*
    6. Disaster Management – *To be taught in First year as a part of syllabus*
    7. Exercise physiology
    8. Dissertation
    9. Practical / clinical examination
2. General design for specialties
    1. Clinical and functional diagnosis in specialty
    2. Concepts of specialty
3. Recent advances in the specialty SPECIALITY OFFERED
    1. Master of Physiotherapy in Musculoskeletal science
    2. Master of Physiotherapy in Neuroscience
    3. Master of Physiotherapy in Cardio-Pulmonary science
    4. Master of Physiotherapy in Sports science
    5. Master of Physiotherapy in Pediatrics and neonatal sciences
    6. Master of Physiotherapy in Obstetrics and Gynecology science
    7. Master of Physiotherapy in Oncology science
    8. Master of Physiotherapy in Community Rehabilation

1.20. M.P.T. Curriculum

1.20.1. COURSE CODE -M.P.T-101

COURSE TITLE - **Laws, Ethics & Administration and Eductional Methodology: (LEM)**

Course Contents: M.P.T LEM Theory (L)

SECTION -A: ETHICS AND LAW

LEM 1.1. Principles of ethics History and evolution of ethics - Helsinki declaration; Nuremberg Code; Principles of ethics and its importance - Autonomy, Beneficence, Non-maleficence, Justice

LEM 1.2. Professionalism

LEM 1.3. Ethics in professional practice Principles of practice in respective profession. Privacy, confidentiality, shared decision making, informed consent, equality and equity, justice

LEM 1.4. ICMR Guidelines General principles, Responsible conduct of research, Risk benefit assessment

LEM 1.5. Informed Consent Process Components of informed consent document, Procedure in obtaining informed consent, Special situations, waivers, and proxy consent

LEM 1.6. Roles and Responsibilities of IEC Ethical Review process, Classification of projects for review, Roles and responsibilities of members, Communications with investigators and authorities

LEM 1.7. Ethics in Special and Vulnerable Populations Types of Vulnerability and vulnerable population, Challenges for research in vulnerable population, Guidelines for research in special and vulnerable population

LEM 1.8. Conflict of Interest Definition and Types of Conflict of Interest, Identifying, mitigating and managing Conflict of Interest, Conflicts of interest in international collaborations

LEM 1.9. Publication Ethics Importance of publishing, Authorship guidelines according to ICMJE, Plagiarism

LEM 1.10. Laws governing Physiotherapy practice: NCAHP Act, Consumer Protection Act, Rights of persons with disability act Ethical issues in practice of Physiotherapy-Clinical, Research and Academics

SECTION -B: Management and administration in Physiotherapy

LEM 2.1. Principles and applications of Management and Administration to Physio Therapy practice:

LEM 2.2. Management PROCESS: planning, organizing, staffing, finance, marketing, controlling, directing.

LEM 2.3. Quality assurance: Total Quality Management: basis of quality management, qualityassurance program in hospitals, medical audit and international quality system.

LEM 2.4. COMMUNICATION: Process of Communication Barriers to Communication Types of Communication Written vs. Oral Communication Elements of good communication

LEM 2.5. Hospital as an organization: functions and types of hospitals MANAGEMENT IN HOSPITAL Setting of a physiotherapy service unit

SECTION-C: Management of Teaching Institution and Educational Methodology In Physiotherapy

LEM 3.1. Education: definition, aims and objectives of education, Agencies of education, Formal and informal education, brief introduction to the philosophies of education, taxonomy of educational objectives, essentials of Physiotherapy education, NEP

LEM 3.2. Basics of Adult Learning Theories including Learning Styles and Motivation

LEM 3.3. Concept of teaching – learning - nature of learning, type and stages of learning, factors affecting learning, laws of learning, learning style teaching learning process, role of teacher in teaching learning process, Adult learning

LEM 3.4. Teaching skills, Teaching Methods in Classroom Setting, clinical teaching methods, planning of teaching: lesson planning and unit planning Teaching aids and educational technology

LEM 3.5. Formulating Intended Learning Outcomes Including Tyler's principles, Bloom's Taxonomy, Miller's Pyramid, Clinical Competence, and Dreyfus' Model of Skill Acquisition

LEM 3.6. **Entrepreneurship in Physiotherapy Practice: Need, Advantages and Opportunities,**

Recommended books for LEM

1. Beauchamp and Childress, Principles of Biomedical Ethics, Fourth Edition. Oxford.
2. Patricia A Marshall. Ethical challenges in study design and informed consent for health research in resource poor settings. World Health Organization. 2007.
3. Natio4n2a6l Ethical guidelines for Biomedical and Health Research involving human participants. Indian Council of Medical Research. 2017.
4. ABC of Learning and Teaching in Medicine. Editor(s): Peter Cantillon, Diana Wood, Sarah Yardley. Ed: 3
5. Understanding Medical Education: Evidence, Theory, and Practice, Editor(s): Tim Swanwick Kirsty Forrest Bridget C. O'Brien. Ed 3
6. Principles of Medical Education. Editor(s): Tejinder Singh, Piyush Gupta, Daljit Singh. Jaypee Brothers. 2012. New Delhi.

1.20.2. COURSE CODE -M.P.T-102

COURSE TITLE - **Research methodology and Biostatistics and Evidence based practice (RMB)**

Course Contents: M.P.T RMB Theory (L)

SECTION-A: RESEARCH METHODOLOGY

RMB 1.1. Introduction to research

RMB 1.2. Types of research

RMB 1.3. Defining a research question

RMB 1.4. Qualitative study designs

RMB 1.5. Quantitative study

RMB 1.6. Type I and type II bias

RMB 1.7. Study design: types

RMB 1.8. Case study, Case series, longitudinal cohort, Pre post design, Time series design, repeated measures design, Randomized control design.

RMB 1.9. Sampling design, calculating minimum sample size based on design

RMB 1.10. Measurement: Properties of measurement: reliability, validity, responsiveness, MCID.

RMB 1.11. Outcome measures: Use of outcome measures in rehabilitation research

RMB 1.12. Research Methods: Designing methodology, Reporting results, Type I and Type II bias.

RMB 1.13. Communicating research.

RMB 1.14. Evaluating published research: looking at the evidence

RMB 1.15. Introduction to evidence-based practice, evaluating evidence,

RMB 1.16. Asking clinical questions

RMB 1.17. Translating of evidence into practice: strategies

RMB 1.18. Use of clinical practice guidelines, clinical pathways, prediction rules to inform practice.

SECTION-B: BIOSTATISTICS

RMB 2.1. Descriptive Statistics and measurement variability

RMB 2.2. Inferential Statistics

RMB 2.3. Comparison of group means: T-test

RMB 2.4. Analysis of variance

RMB 2.5. Multiple comparison tests

RMB 2.6. Parametric and Non parametric tests

RMB 2.7. Correlations

RMB 2.8. Regression

RMB 2.9. Analysis of frequencies: Chi square

RMB 2.10. Statistical measure of validity and reliability

RMB 2.11. Factorial Design analysis

RMB 2.12. Power analysis – Determining sample size, Epidemiological Measures – Rate, Ratio, Proportion, Incidence and prevalence, Relative risk, Risk ratio, Odds ratio

RMB 2.13. Application of various statistical software.

SECTION-C: SCIENTIFIC WRITING

RMB 3.1. Definition and kinds of scientific documents – Research paper, Review paper, Book, Reviews, Thesis, Conference and project reports (for the scientific community and for funding agencies).

RMB 3.2. Publication – Role of author, Guide, Co-authors.

RMB 3.3. Structure, Style and contents; Style manuals (APA, MLA); Citation styles: Footnotes, References; Evaluation of research

RMB 3.4. Significance of Report writing; Different steps in Report writing; Mechanics and precautions of writing research reports Oral and poster presentation of research papers in conferences/symposia; Preparation of abstracts.

RMB 3.5. Structure of Thesis and Content – Preparing Abstracts.

Recommended books for RMB

1. Bailey, N.T.J. -Statistical methods in Biology. The English universities press, London
2. Bajpai, S.R.- Methods of Social Survey and Research, Kitab Ghar, Kanpur.
3. Colton - Statistics in medicine, Little Brown Company, Boston
4. Gupta, S.P -Statistical methods. Sultan Chand and Sons Publishers , New Delhi.
5. Goulden C.H.- Methods of Statistical Analysis. Asia Publishing House , New Delhi.
6. Mohsin S.M.- Research Methods in Behavioral Sciences: Orient Publications. New Delhi.
7. Mahajan - Methods in Biostatistics, Jay Pee Brothers.Medical Publishers (P) Ltd. New Delhi.
8. Hicks- Research for Physiotherapists, Churchill Livingstone, London.
9. Meenakshi. - First Course in Methodology of Research. Kalia Prakashan, Patiala.
10. Kumar , R.- Research Methodology. Pearson Education , Australia.
11. Snedecor,G.W -Statistical Methods, Allied Pacific Pvt. Ltd., London
12. Singh, I.- Elementary Statistics for Medical Workers. Jaypee Brothers Medical Publishers (P) Ltd. New Delhi.
13. Rehabilitation Research: Principles and Applications by Elizabeth Domholdt (Elsevier Science Health Science Div, 2004)

1.20.3. COURSE CODE -M.P.T-103

COURSE TITLE -BIOMECHANICS & THERAPEUTICS (BCT)

Course Contents: M.P.T BCT Theory (L)

SECTION A – Concepts of Biomechanics:

BCT 1.1. Introduction to Kinesiology and Biomechanics. Biomechanics of Tissues and structures of the musculoskeletal system

BCT 1.2. Principle of Biomechanics

BCT 1.3. Nature and importance of Biomechanics in Physiotherapy.

BCT 1.4. Methods of kinetics and kinematics investigation

BCT 1.5. Introduction to biomechanical analysis of human motion.

BCT 1.6. Analytical tools and techniques –

1.20.3.1. Isokinetic Dynamometer,

1.20.3.2. Kinesiological EMG,

1.20.3.3. Electronic Goniometer,

1.20.3.4. Force Platform,

1.20.3.5. Videography.

BCT 1.7. Upper Extremity: Shoulder and Shoulder girdle, Elbow joint, Wrist joint and Hand.

BCT 1.8. Lower Extremity: Pelvic Girdle, Hip joint, Knee joint, Ankle & Foot

BCT 1.9. Spine

BCT 1.10. Gait

BCT 1.11. Gait Analysis: Kinetic & Kinematic Analysis.

BCT 1.12. Pathological Gait: Kinetic & Kinematic Analysis

BCT 1.13. Ergonomic approach to lifting and handling, workspace and environment. Patient positioning, body mechanics and Transfer techniques

SECTION-B: Physiotherapy techniques

BCT 2.1. Principle of therapeutic exercises

BCT 2.2. Definition, details of effects and uses of following exercises.

BCT 2.3. Dynamic Exercises

BCT 2.4. Plyometric Exercises

BCT 2.5. Isokinetic Exercises

BCT 2.6. Kinetic chain exercises

BCT 2.7. Balance and coordination exercises

BCT 2.8. Biophysics of contractile and non-contractile tissues, Response to mechanical loading

BCT 2.9. Clinical reasoning and differential clinical diagnosis based on various approaches such as Maitland, Kaltenborne, Cyriax, Mulligan, Mckenzie etc.

BCT 2.10. Proprioceptive neuromuscular Facilitation,

BCT 2.11. Hydrotherapy Techniques

BCT 2.12. Functional assessment and re-education

BCT 2.13. Yoga: Introduction, Historical background and Origin of Yoga, Meaning and Concept of Yoga and its relationship with Physical Education and Sports, **Yoga in Global Scenario**, **Pranayama:** Meaning, Types and its importance. **Asanas:** Asanas- meaning, types, principles, Techniques of asanas and effects of asanas on various systems of the body - circulatory, respiratory and digestive system.

BCT 2.14. Electro diagnosis: introduction to methods of electro diagnosis SD CURVE

BCT 2.15. Electromyography: technique of EMG, interpretation of normal and abnormal responses

BCT 2.16. Nerve conduction studies: MNCV, SNCV, variables affecting nerve conduction, measurement of NCV of nerves of upper limb and lower limb, interpretations of normal and abnormal responses.

BCT 2.17. Evoked potentials, H-reflex, P wave, repetitive nerve stimulation, VEP, BAEP, SSEP, SSR.

BCT 2.18. Review of Principles underlying the application of following modalities with reference to their Production, biophysical and therapeutic effects, indications and contraindications and the specific uses of:

1. Superficial heating modalities
2. Deep heating modalities
3. Ultrasound
4. Cryotherapy

BCT 2.19. Review of Principles underlying the application of following modalities with reference to their Production, biophysical and therapeutic effects, indications and contraindications and the specific uses of Physiotherapy

BCT 2.20. Low Frequency Current: Diadynamic Current, Iontophoresis

BCT 2.21. High Voltage, Pulsed Galvanic Stimulation, TENS, IFT, Russian Currents. LASER

BCT 2.22. Advanced Electro Therapeutics in Tissue healing, Wound care, Management of Scars, keloids, Muscle Plasticity & Integumentary Conditions.

BCT 2.23. BIO-FEED BACK

Recommended books for BCT

1. James G. Hay – The Biomechanics of Sports Techniques, Prentice Hall.
2. Brunnstrom - Clinical Kinesiology, F.A. Davis.
3. Luttgens K., Hamilton N.: Kinesiology – Scientific Basis of Human Motion, Brown & Benchmark.
4. Kreighbaum E., Barthels K.: Biomechanics – A Qualitative approach for studying human Motion, MacMillan.
5. Rasch and Burk: Kinesiology and Applied Anatomy, Lee and Fabiger.
6. White and Punjabi - Biomechanics of Spine - Lippincott.
7. Norkin & Levangie: Joint Structure and Function - A Comprehensive Analysis - F.A.
8. Davis.
9. Kapandji: Physiology of Joints Vol. I, II & III, W.B. Saunders.
10. Northrip et al: Analysis of Sports Motion: Anatomic and Biomechanics perspectives,
11. W.C. Brown Co., IOWA.
12. Leveac B.F.: Basic Biomechanics in Sports and Orthopedic Therapy, C.V. Mosby.
13. De Boer & Groot: Biomechanics of Sports, CRL Press, Florida.
14. Basmajian - Muscle alive - Williams & Wilkins.
15. Nordin & Frankel - Basic Biomechanics of Muscular Skeletal Systm - Williams & Wilkins.
16. Bartlet - Introduction to Sports biomechanics - F & FN Spon Madras.

1.20.4. Locomotor disability Assessment content:

DISABILITY (PERMANENT PHYSICAL IMPAIRMENT) ASSESSMENT AND CERTIFICATION GUIDELINES & GAZETTE NOTIFICATION:

Detail study of Government Gazette to be done: (The Gazette of India is regularly updated, and its publications can change over time. Refer the recent Gazette publications issued by the Government of India, from the official website)

PWD Act 1995 and Rights of person with Disability Act 2016, **to study in detail.**

1.20.5. BLS and ACLS Training:

Course Title: Basics of Emergency Care and Life Support Skills (ECLS): Theory (L) Practical (P)

ECLS 1.0. Subject Description and instruction to teacher

Basic life support (BLS) is the foundation for saving lives following cardiac arrest. Fundamental aspects of BLS include immediate recognition of sudden cardiac arrest (SCA) and activation of the emergency response system, early cardiopulmonary resuscitation (CPR), and rapid defibrillation with an automated external defibrillator (AED). Initial recognition and response to heart attack and stroke are also considered part of BLS. The student is also expected to learn about basic emergency care including first aid and triage. The purpose of this course is to equip the students with the skill to save the life of a person in different emergency situation as first responder. The training should be provided using Mannequins and dummies and Videos presentations and Role plays should be also used to impart knowledge and skill besides the lecture - demonstrations.

ECLS 1.0.1. Course Outcomes:

After completion of this course the student shall be able to

1.20.5.1. Perform Opening and maintaining and patent airway: assessment and knowledge of airway maneuvers and adjuncts

1.20.5.2. Ventilate patients: Assessment and management of breathing with Mouth to mouth and mouth to mask

1.20.5.3. Administer basic life support skills including cardiopulmonary resuscitation

1.20.5.4. Provide first aid of simple and multiple system trauma such as • Controlling hemorrhage • Managing Burns and wounds • Response to effects of weapons of mass destruction • manually stabilizing injured extremities

1.20.5.5. Provide first aid to patients with medical emergencies like heart attack and stroke • Identifying signs of Stroke and heart attack and safe transfer after first aid without delay in transfer. • Manage general medical complaints seizures and animal bites (snake /dog bite)

1.20.5.6. Reassure patients and bystanders by working in a confident, efficient manner • Avoid mishandling and undue haste while working expeditiously to accomplish the task

1.20.5.7. Manage safe patient transport Entailing-Extrication of the victim, helmet removal and spine protection during transport.

1.20.5.8. Explain Roles, responsibilities and limitation of first responder.

Course Contents:

SECTION -A

UNIT 1

ECLS 1.1. Emergent conditions and magnitude, Concept of golden hour, Duties and responsibilities of first responder

ECLS 1.2. Ethical issues and Gather information from observation, experience and reasoning. Identification of rapidly changing situations and adapt accordingly. Planning and organization of work. Scene safety. Dealing with emotional reactions family members and bystanders

ECLS 1.3. Well-being of first responder Personal protection

1. Steps to be taken against airborne and blood-borne pathogens
2. Personal protective equipment necessary for each of the following situations: Hazardous materials Rescue Operations Violent Scenes Crime scenes
3. Electricity, Water and ice
4. Exposure to blood-borne pathogens Exposure to airborne pathogens

UNIT 2

ECLS 2.1. Airway

1. Signs of inadequate breathing
2. Mechanism of injury to opening the airway
3. Steps in the head-tilt chin-lift
4. Steps in the jaw thrust
5. Taking out foreign body
6. Ensuring patent airway during seizures and vomiting.

ECLS 2.2. Ventilation

1. Of a patient with a mask or barrier device
2. Steps in providing mouth-to-mouth and mouth-to-stoma ventilation

ECLS 2.3. Circulation

1. Evaluate the cardiac status of the patient
2. Determine the need for and take necessary action to proper circulation
3. Steps for control of bleeding: Pressure bandage and tourniquet

ECLS 2.4. Clearing a foreign body airway obstruction

ECLS 2.5. CPR

1. Implications of cardiac arrest
2. Cardiopulmonary resuscitation (CPR)
    i. How it works
    ii. Steps
    iii. When to stop CPR
3. Brief overview of AED Automated external defibrillator (only demonstration –no hands on)

SECTION -B

UNIT 3

ECLS 3.1. Bleeding and Soft Tissue Injuries

1. Difference between arterial and venous bleeding
2. Stopping external bleeding
3. Identification of Internal bleeding
4. types and Functions of dressings and bandages
5. How to help a victim of burn injury

ECLS 3.2. Injuries to Muscles and Bones

1. Suspecting bony/spinal injury
2. Splinting –materials used
3. Importance of splinting

UNIT 4

ECLS 4.1 Medical Emergencies

ECLS 4.2 Identification of the patient steps in providing first aid to a patient with

i. A general medical complaint –

ii. Seizures

iii. Chest-pain

iv. Shortness of breath

v. Exposure to heat

vi. Including other medical complaints like allergy, diarrhea, fainting, low blood sugar, stroke

vii. Drowning

viii. Poisoning

ECLS 4.3 Transportation Importance of timely and proper transportation methods of transportation of victim from site of injury to ambulance. Importance of spine protection methods of spine protection during transportation

ECLS 4.4 Disaster preparedness -. Preparedness and risk reduction Incident command and institutional mechanisms Resource management

Practicals

Student should practice on Mannequins and dummies and should be able to

ECLS (P) 5.1. Provide Airway & Ventilation

ECLS (P) 5.2. Perform Basic Life Support: CPR

ECLS (P) 5.3. Perform Initial management of Thermal injury, electric injury

ECLS (P) 5.4. Perform stabilizing injured extremity and wound management

ECLS (P) 5.5. Demonstrate bandaging techniques to various body parts

ECLS (P) 5.6. Demonstrate Extrication, Helmet removal and spine protection

ECLS (P) 5.7. Demonstrate Stretcher use

Recommended text books for ECLS

Indian red cross : INDIAN FIRST AID MANUAL 2016 (7th edition) available at https://www.indianredcross.org/publications/FA-manual.pdf

1.20.6. Disaster Management:

Course Title: Disaster Management (DM): Theory (L)

DM 1.0 Subject Description and instruction to teacher: The commission's goal is to emphasize the vital role physical therapists (physios) play in disaster management and contribute to national and global preparedness. To achieve this, it's essential to raise awareness among physiotherapists about national and international organizations and emphasize the crucial role physical therapists play in disaster management, particularly within Emergency Medical Teams. Also it may be noted that the acts, policies, gazettes are regularly updated, and its publications can change over time. The teachers and students should thus refer the recent publications issued on the official website

DM 1.0.1. Course Outcomes: After completion of this course the student shall be able to

1.20.6.1. Understand the crucial role physical therapists play in disaster management, particularly within Emergency Medical Teams.

1.20.6.2. Should be able to identify national and international organizations that play a vital role in disaster management

1.20.6.3. Should be able to identify the legal framework for disaster management in India and disaster prone areas.

1.20.6.4. Provide essential information to other physical therapists interested in disaster response work and to make them aware of national and international agencies already active in the field.

1.20.6.5. Promote global preparedness and support physical therapists in making a meaningful difference in disaster response and recovery efforts

Course Content: Disaster Management (DM): Theory (L)

DM 1.1. Definition of disaster and the hazards associated with disaster, Vulnerable groups in Disaster

DM 1.2. Definition of Advocacy, disability advocacy, Contingency planning wrt to disaster management, Hazard, Risk , Vulnerable groups

DM 1.3. History of involvement of Physiotherapists in rehabilitation efforts during emergencies

DM 1.4. National organisations who are involved in disaster preparedness and management strategies:

1. The legal framework for disaster management in India: Key takeaways of Disaster Management Act 2005, National Policy on Disaster Management 2009 and National Disaster Management Plan 2018
2. Different types of disasters managed in India, Epidemiologic surveillance and disease control, main goal of the National Disaster Management Authority, areas in India are most prone to disasters, Institutional structure for disaster management in India at various levels, Central Ministry that coordinates disaster management and leader of NDMA in India
3. Disaster Management Act of 2005 key take aways and its significance, Phases of Disaster management, Long term prevention measures, role of various stake holders in disaster management, role of community involvement in disaster management, challenges faced in disaster management in India
4. Prime minister’s 10 point agenda and Community based and Technology driven approaches: Key policies and strategies

DM 1.5. International organisations who facilitate contributions of physiotherapists in disaster preparedness and management strategies. Role of physiotherapists in:

1. Disaster management within their own countries, benefits of rehabilitation provided following disasters
2. Prevention of a disaster
3. Preparedness for disaster with respect to essential locally appropriate preparedness for a disaster,
4. Identifying and connecting professional associations, health service providers and training institutions.
5. Developing international humanitarian response
6. Response to disaster: Required skills and knowledge and required actions and secure resources with respect to assessment, coordination, psycho-social support and advocacy
7. Recovery: with respect to planning of medical management and local capacity building and physiotherapy rehabilitation, advocacy

DM 1.6. The type and distribution of injuries caused by disasters, the type of hazards, common injuries that can lead to long-lasting or permanent disability.

DM 1.7. Clinical Practice in Response phase along with documentation (conservative and surgical), record management, data and research, informed consent and confidentiality, regulations and scope of practice, hand hygiene and infection control, communication, referral, discharge planning with respect to international management strategies.

DM 1.8. International Disaster Management Rehabilitation Response Plans and role of Physiotherapists with respect to: Systems in Place, Identifying Personnel, Facilities and Resources, Advocacy and Partnerships, Training and Capacity Building

DM 1.9. Elements to be considered "essential" components in any disaster education or training programme for health professionals as defined by Global Response Framework,

DM 1.10. The World Health Organization (WHO) : the lead UN agency in the health cluster and its emergency response framework and Humanitarian principles

Recommended websites for references: Disaster management

National Disaster Management Plan, 2016. A publication of the National Disaster Management Authority, Government of India. May 2016, New Delhi at www.mha.gov.in

www.wcpt.org/disaster-management.

1.20.7. Exercise Physiology

Details presented on next page

1.20.8. **Dissertation:**

Each candidate will have to carry out of a dissertation on Speciality related subject of MPT. Ethical approval certificate from **Registered Institutional Ethical committee** and Clinical Trial Registration is mandatory for intervetional Dissertation study topic. The dissertation to be guided by Guide of the speciality of faculty of physiotherapy of the department under whom the student is persuing MPT. The dissertation will be evaluated by the External/Internal Examiners. The final dissertation duly approved by the External/Internal examiners will be submitted to the Dean/Principals office with the result. The dean/ Principal's office will send the dissertation to the library for record.

1.20.9. **Practical / clinical examination**

Compulsary rotatory Clinical Posting as per the Speciality and Clinical Assessment during Clinical posting is mandatory .

2ND YEAR M.P.T

1.20.10. COURSE CODE -M.P.T-201

COURSE TITLE -EXERCISE PHYSIOLOGY (EP) Theory (L) Practical (P)

EP 1.0. Subject description Course outcomes

1. CO1: Comprehend the basic knowledge of sources of energy, aerobic and anaerobic synthesis of ATP along with the understanding of utilization of substrates in relation to the intensity and duration of exercise
2. CO2: Appreciate the measurement of energy cost of exercise and importance of energy transfer and energy expenditure at rest and during various physical activities
3. CO3: Understand the role of various macro and micro nutrients as well as their caloric requirements along with the basic classification, functions and utilization of nutrients.
4. CO4: Acquire about importance of diet for various competitions, nutrient supplements for performance and to design caloric requirements for various sports and age groups.
5. CO5: Critically evaluate the central and peripheral mechanism that regulates the cardiovascular and respiratory systems in exercise along with the physiological responses and adaptations of these systems to exercise and training.
6. CO6: Identify the regulation and significance of acid base balance following exercise CO7: Understand the responses of various hormones with respect to exercise

SECTION -A

EP 1.1. Bioenergetics of exercise: High energy phosphates, Anaerobic and aerobic ATP synthesis, Bioenergetics Control, exercise intensity & substrate utilization, protecting CHO stores, muscle adaptation to endurance training, processes that potentially limit the rate of fat oxidation, regulation of substrate utilization, training - induced increase in FFA oxidization:

EP 1.2. Basal metabolic and resting metabolic rates and factors affecting them, Classification of Physical Activities by energy expenditure. Concept of MET measurement of energy cost of exercise

EP 1.3. **Nutrition metabolism** of Carbohydrate, fats, proteins, vitamin, mineral and water

EP 1.4. **Nutrition in exercise** optimum nutrition for exercise, nutrition for physical performance, pre game meal carbohydrate loading, food for various athletic events, fluid and energy replacement in prolonged exercise

EP 1.5. **Respiratory responses to exercise:** Ventilation at Rest and during Exercise, Ventilation and the Anaerobic Threshold, static and dynamic lung volume. Gas diffusion, Oxygen and carbon dioxide transport second wind, stich by side control of pulmonary ventilation during exercise adaptive changes in the respiratory systems due to regular physical activities.

EP 1.6. Cardiovascular responses to exercise- Cardiovascular system and exercise, acute vascular effects of exercise, Circulatory responses to various types of exercise regulation of cardiovascular system during exercise, Pattern of redistribution of blood flow during exercise, adaptive responses of cardiovascular system to aerobic and anaerobic training. Athlete heart

EP 1.7. Exercise and Acid Base Balance: Acid and Bases, Buffers, pH, Respiratory Regulation of pH, Alkali Reserve, The kidneys and Acid base balance, Alkalosis and Acidosis, Acid base balance following heavy exercise.

EP 1.8. Hormonal responses to exercise with respect to Growth Hormone (GH), Thyroid and Parathyroid Hormones. Antidiuretic Hormone (ADH) and Aldosterone, Insulin and Glucagons, The catecholamine; epinephrine and norepinephrine. The sex hormones. The glucocorticoids (Cortisol) and Adreno Corticotropic Hormones (ACTH). Prostaglandins and Endorphins.

SECTION -B

EP 2.1. Training and conditioning

Physiological basis of physical training, training principles, interval training, continues running concept of anaerobic threshold and vo2 max, physiological effects of various physical training methods- aerobic and anaerobic training, strength training factors influencing training effects – intensity, frequency, duration, detraining, process of recovery, post exercise oxygen consumption factors affecting recovery process, overtraining

EP 2.2. Body temperature regulation during exercise

Mechanism of regulation of body temperature, Body temperature responses during exercise, Physiological responses to exercise in the heat, Acclimatization to exercise in the heat, Effects of age and gender on body temperature regulation during exercise, Physical activity and heat illness [heat exhaustion, dehydration exhaustion heat cramps & heat stroke] Prevention of Heat Disorders

EP 2.3. Exercise in the Cold

Effects of exposure to cold and severe cold, Wind chill, Temperature receptors, Role of hypothalamus, shivering, Frost Bite and other problems, Clothing and Environment

EP 2.4. Exercise at Altitude

Exercise at altitude immediate physiological responses at high altitude, physiological basis of altitude training, phases of altitude training and specific training effects, altitude acclimatization, oxygen dissociation curve at altitude, disorders associated with altitude training

EP 2.5. Exercise and body fluids

Measurement and regulation of body fluids, Body fluid responses and adaptations to exercise, Effects of dehydration and fluid replenishment on physiological responses to exercise and performance Fluid/carbohydrate replacement beverages

EP 2.6. Physical activity, body composition, energy balance and weight control

Significance and measurement of body composition, Body composition during growth and aging, Body composition and physical performance, Effect of diet and exercise on body composition, Physical activity, energy balance, nutrient balance and weight control, Physical activity, fat distribution and the metabolic syndrome , Healthy weight loss, Ways and methods of weight reduction , fluid maintenance, disordered eating, nutritional ergogenic aids, diet supplements in athletes and others involved in physical activity.

EP 2.7. Exercise and Diabetes Mellitus

Exercise in insulin, requiring diabetes and non-insulin dependent diabetes mellitus, Effect of physical training on glucose tolerance and insulin sensitivity, Management of diabetes by diet and insulin

Books suggested for EP

1. Essentials of Exercise Physiology: McArdle, WD, Katch, FI, and Katch, VL. Lippincott Williams and Wilkins.
2. Fundamentals of Exercise Physiology: For Fitness Performance and Health, Robergs RA, and Roberts, S.O. McGraw Hill
3. Exercise Physiology: Powers, SK and Howley ET; Mc Graw Hill
4. Physiology of Sport and Exercise: Wilmore, JH and Costil, DL. Human Kinetics
5. Exercise Physiology- Human Bioenergetics and its Application: Brooks, GA, Fahey, TD, White, TP. Mayfield Publishing Company
6. Komi, P. (Ed.) Strength and power in sport. Blackwell Scientific Publications.
7. Levick, J.R. An introduction to Cardiovascular Physiology. 2nd ed. Butterworth Heinemann
8. McArdle, WD, Katch, FI & Katch, VL Exercise Physiology. Lippincott, Williams & Wilkins.
9. Shephard and Astrand Endurance in sport. Blackwell Scientific Publications.
10. Willmore, JH & Costill, DL Physiology of Sport and Exercise. 2nd ed. Human Kinetics.
11. Guyton, A.C. Textbook of Medical Physiology. Philadelphia: Saunders,
12. Nutrition for sport and exercise; Berning and Steen

1.20.11. Specialty papers

COURSE CODE -M.P.T-104, M.P.T 202, & M.P.T-203

1) Master of Physiotherapy in Musculoskeletal Sciences

MPT (MS)104: Clinical, Physical and Functional diagnosis in Musculoskeletal Physiotherapy

MPT (MS) 202: Musculoskeletal Physiotherapy

MPT (MS) 203: Recent advances in Musculoskeletal Physiotherapy

COURSE CODE-M.P.T (MS)-104

COURSE TITLE **Clinical, Physical and Functional diagnosis in Musculoskeletal physiotherapy (MCPFD)**

MCPFD 1.0. Subject description

MCPFD 1.0.1. Course outcome students will be able to:

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in Orthopedics& interpret clinical tests and special investigations commonly used in the diagnosis of these conditions.
2. Generate a primary diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on Orthopedic Physiotherapy practice.
5. Demonstrate a broad range of technical skill in diagnosing the Physiotherapy related Orthopedic conditions.
    a) Cardiac efficiency tests and spirometry
    b) Fitness test for sports
    c) Physical disability evaluation and disability diagnosis. Gait analysis and diagnosis.
    d) Coping Strategies in chronic painful musculoskeletal conditions. Checkouts of orthotics and prosthetics for neuro-musculoskeletal problems. Effect of Immobilization on Musculoskeletal System
    e) Application of ICF in Musculoskeletal diagnosis
    f) Medical screening for potential referred pain and Red Flags

Course Content: M.P.T (MS)-104

Part I

MCPFD 1.1. Clinical Decision Making - Planning Effective Treatment. Clinical decision making models, Team approach, Foundation for clinical decision making.

MCPFD 1.2. Vital Signs. Identification of reasons for monitoring vital signs; importance of monitoring vital signs; common techniques of monitoring vital signs; identification and analysis of normal values with that of abnormal values.

MCPFD 1.3. Principles and application of investigative and imaging techniques in Physiotherapy

a. Blood test

b. Arterial Blood Gas (ABG) analysis

c. Pulmonary Function Test (PFT)

d. Radiological examination

e. Computerized Tomography (CT)

f. Magnetic Resonance Imaging (MRI)

g. Ultrasonography (US)

h. Electrocardiography (ECG)

i. Dope testing

MCPFD 1.4. Evaluation assessment and treatment planning strategies for musculoskeletal, neurological, cardiopulmonary, sports specific and other physiotherapy conditions: Principles of evaluation, clinical manifestations, general and specific clinical examination.

i. Physiotherapy assessment of the following:

a. Range of motion (ROM)

b. Tone

c. Muscular strength and endurance

d. Flexibility

e. Coordination - Non equilibrium test - Equilibrium test

f. Sports specific skills

g. Cardiac efficiency

h. Sensory evaluation

i. Functional Evaluation - Various scoring methods in functional assessment - Validity and reliability

j. Fitness evaluation - Aerobic - Anaerobic

k. Spasm

l. Trigger Point

m. Tender Point

n. Spasm

ii. Assessment of cognitive, perceptual dysfunctions and vestibular dysfunction.

MCPFD 1.5. Electro-Diagnosis:

i. Characteristics and components of Electro therapeutic stimulation systems and Electro physiological assessment devices.

ii. Instrumentation for neuromuscular electrical stimulation.

iii. Electrical properties of muscle and nerve.

iv. Neurobiology of afferent pain transmission and central nervous system mechanisms of pain modulation.

v. Electrical stimulation and circulation.

MCPFD 1.6. Clinical Electro physiological testing: Instruments, Techniques and Interpretations of

a. Nerve conduction velocity including Repetitive Nerve Stimulation (RNS)

b. Electromyography

c. Bio-feedback technique.

d. Late responses

MCPFD 1.7. Concepts of electro physiological studies in neuro muscular diseases as a diagnostic and therapeutic tool.

MCPFD 1.8. Evoked potentials – VEP, SSEP, MEP, BAEP

Part II

MCPFD 2.1. Psychological aspects of rehabilitation in disability: Psychological tests.

MCPFD 2.2. Developmental Screening

i. Factors Motor control assessment

ii. Motor control theories/mechanism

iii. Patterns of normal development

iv. specific procedures and tests used to assess motor control defects

MCPFD 2.3. Anthropometry

1. Body measurements - Height - Weight - Circumference
2. Body Proportion - Body Mass Index (BMI) - Waist Hip Ratio (WHR)

3. Body Composition

   i. Somatotyping

   ii. Methods of measurement

      a. Water displacement method for body fat.

      b. Skin fold measurement

      c. Under water weighing

      d. Bioelectric Impedance Analysis (BIA)

MCPFD 2.4. Differential diagnosis in Physiotherapy

MCPFD 2.5.

i. Functional evaluation.

   a. The concepts of health status impairment; functional limitations;
   b. Disability and Persons with Disabilities;
   c. Definition of functional activity and the purposes and components of the functional assessment;
   d. Selection of activity and roles for an individual based on his or her capabilities and functional limitations.

ii. Various forms of functional tests;

   a. Physical function test
   b. Multi-dimensional functional assessment instrument,
   c. Identification of instrument for testing function.

iii. Various scoring methods used in functional assessment;

iv. Reliability and validity of various functional assessments.

MCPFD 2.6. Evaluation of aging

1.20.11.A. SPECIALITY PAPER 2 COURSE CODE-202

COURSE CODE-M.P.T (MS)-202

COURSE TITLE **Musculoskeletal physiotherapy (MSK)**

MSK 1.0. Subject description

MSK 1.0.1. Course outcome students will be able to:

1. Develop a management plan, generally including some lifestyle factors, in co-operation with the Clinical Supervisor and consider a prognosis that reflects on the patient's problem.
2. Manage a patient in consultation and co-operation with the clinical supervisor, identifying the presenting problem, developing a basic working diagnosis and selecting a treatment regime that considers the presenting problem with some consideration for ethical, practical and pragmatic concerns.
3. Maintain legal (accurate, clear and legible) patient histories, write basic referral letters and recognize the need of further referral in conference with Clinical Supervisor and peers.
4. Discuss the Common exercise prescriptions and their clinical use, and the sequence of treatment and how to advise different sorts of patients

SECTION -A

MSK 1.1. Advanced instruction in treatment and follow-up of the musculoskeletal system

MSK 1.2. Upper Quarter and Lower Quarter Muscle imbalances leading to dysfunction with corrective measures Exercise planning and Exercise Prescription for musculoskeletal conditions

MSK 1.3. Management of pathological gaits and Postural deviations

MSK 1.4. Orthopaedic implants - designs, materials indications, post – operative Physiotherapy

MSK 1.5. Manual therapy – Principles, indications, contraindications, and methods of application of joint mobilization techniques and soft tissue manipulations Cumulative Traumatic Disorders and management

MSK 1.6. Aids and appliances, adaptive functional devices to improve neuro-musculoskeletal dysfunctions Physiotherapy management of locomotor impairments, and disabilities at institutional & community levels

MSK 1.7. Taping techniques in orthopedic conditions Sports injuries and their management

SECTION- B

MSK 2.1. Physiotherapy management in Fractures, Joint Instabilities, Soft Tissue Disorders, Deformities, Nerve Injuries, Metabolic, Hormonal Conditions, Neoplastic, Infective Conditions of Bones and Joints of musculoskeletal system pertaining to upper quarter lower quarter and spine

MSK 2.2. Pre and Post surgical Rehabilitation of Joint replacement surgeries

MSK 2.3. Physiotherapy management after tendon transfer, Electrical stimulation and biofeedback procedures Assessment and management of Paediatric and geriatric musculoskeletal disorders

MSK 2.4. Physical Agents and Electrotherapeutic management in orthopedic conditions. Rehabilitation of congenital conditions and malformation of musculoskeletal disorders. Physiotherapy management in Amputation and Prosthetic Prescription.

MSK 2.5. Equipment in orthopedic Physiotherapy such as: Isokinetic, EMG and Biofeedback, Proprioception assessment equipments, Gait analyzers. Home and self-help programme in orthopedic Physiotherapy.

MSK 2.6. Disability prevention and management

1.20.11.B SPECIALITY PAPER 3

COURSE CODE-M.P.T (MS) 203

Course Title: Recent advances and Evidence Based Practice in Musculoskeletal Physiotherapy (MRAEB)

MRAEB 1.0. Subject description

MRAEB 1.0.1. **Course outcome**

Students will be able to:

1. Understand and apply the information regarding recent advances in Orthopedic Manual Therapy for patient care.
2. Search the evidences available for assessment and management of orthopedic conditions.
3. Apply the evidences available for the management of various orthopedic conditions.

SECTION A:

MRAEB 1.1. Manual therapy: soft tissue manipulations and mobilization, neural mobilization, acupressure.(Cyriax, Maitland, Butler, McKenzie, Kaltenborn, Mulligan)

MRAEB 1.2. EBP and Recent advances in clinical assessment, laboratory investigations and diagnosis of musculoskeletal disorders. EBP in Management of pain in musculoskeletal disorders.

MRAEB 1.3. Recent Advances in management of orthopedic conditions- medical, surgical and Physiotherapy Recent Advances in Physiotherapy management in arthritis and allied conditions.

MRAEB 1.4. Recent Advances and Controversies in Electrotherapy for orthopedic conditions.

MRAEB 1.5. Assessment and training for Core, postural stability and balance in musculoskeletal conditions Recent advances in Kinematic & kinetic analysis.

MRAEB 1.6. Use of advance Assistive devices and technologies in musculoskeletal system Current trends in sports injuries and management.

MRAEB 1.7. Evidence Based Physiotherapy in management of metabolic and hormonal, neoplastic and infective conditions of bones and joints.

SECTION-B

MRAEB 2.1. Recent Advances in Physiotherapy following arthroplasty, implants and soft tissue repairs.

MRAEB 2.2. EBP and recent advances in Physiotherapy after tendon transfer, Electrical stimulation and biofeedback procedures. EBP in Rehabilitation of congenital conditions and malformation of musculoskeletal disorders.

MRAEB 2.3. Recent Advances in External aids, appliances, adaptive self-help devices; prescription, biomechanical compatibility, check- out and training. EBP and Recent advances in electro diagnosis, Electromyography, NCV and evoked potential studies.

MRAEB 2.4. Community based rehabilitation in musculoskeletal disorders. Recent Advances and Controversies in Orthopaedic physiotherapy. Ergonomics assessment and management at work place.

MRAEB 2.5. Evidence Based Practice and Recent Advances of Manual Therapy in Musculoskeletal Conditions Evidence based practice and recent advances of Aquatic therapy in Orthopaedic conditions.

Suggested reading

1. Jones, M. A., &Rivett, D. A. Clinical reasoning for manual therapists. Edinburgh: Butterworth Heinemann.
2. Eyal Lederman - Fundamentals of manual therapy.
3. Grieve's Modern manual therapy
4. Walter Herzog - Clinical Biomechanics of spinal manipulation
5. Sandy Fritz, Kathleen Paholsky and M.JanesGrosenbach - Basic Science for soft tissue and movement therapies.
6. Jean Sayne Adams, Steve Wright - Theory and practice of therapeutic touch.
7. AkhouryGourang Sinha – Principle and practice of therapeutic massage
8. Carol Manheim – The Myofascial release manual 3rd Edition
9. Maitland's – Peripheral manipulation
10. Maitland's – Vertebral manipulation
11. Chaitow – Cranial manipulation theory and practice
12. Lynn Paul Taylor – Taylor's manual of physical evaluation and treatment
13. Denise Deic – Positional release technique from a dynamic systems perspective.
14. Goodman and Snyder – Differential diagnosis in physical therapy
15. Tidy's Physiotherapy, Elsevier Publication.
16. Chaitow - Muscle energy technique
17. Reid et al – Sports injury assessment and rehabilitation.
18. Kjaer et al – Text book of sports medicine
19. Scudder Mc Can - Sports medicine, A comprehensive approach
20. Norris – Sports injuries, diagnosis and management for physiotherapists.
21. Werner Kuprian – Physical therapy for sports.
22. McGinnis – Biomechanics of sports and exercises.
23. Chew, F. Skeletal radiology: The bare bones. Baltimore, MD: Williams & Wilkins.
24. Eisenberg, R. L., & Johnson, N. M. Comprehensive radiographic pathology St Louis, MO: Mosby.
25. Hughes, J., & Hughes, M.. Imaging: Picture tests. Edinburgh: Churchill Livingstone.
26. Mace, J. D., & Kowalczyk, N. Radiographic pathology for technologists. St Louis, MO: Mosby.
27. Redhead, D. N. Imaging: Colour guide. Edinburgh: Churchill Livingstone.

28. Yochum, T. R., & Rowe, L. R. Yochum and Rowe's essentials of skeletal radiology. Baltimore, MD: Lippincott Williams & Wilkins.
29. Gunn, C. Bones and joints: A guide for students. London: Churchill Livingstone.
30. Haines, D. E. Fundamental neuroscience W. B. Saunders Co.
31. Kandel, E. R., Schwartz, J. H., & Jessell, T. M. Principles of neural science McGraw-Hill.
32. Longmore, J., Wilkinson, I., & Rajagopalan, S. Oxford handbook of clinical medicine Oxford: OUP.
33. Newman D4o4r5land, W. A. Dorland's illustrated medical dictionary W. B. Saunders Co.
34. Nolte, J. The human brain: An introduction to its functional anatomy. St Louis, MO: Mosby.
35. Nolte, J., & Angevine, Jr. J. B. The human brain in photographs and diagrams. St Louis, Mosby.
36. Wicke, L. Atlas of radiologic anatomy, Munich, Germany: Lea &Febiger.
37. Seidel, H. Mosby's guide to physical examination. St Louis, MO: C.V. Mosby.
38. Cailliet, R. Neck and arm pain Philadelphia: FA Davis.
39. Cailliet, R. Shoulder pain Philadelphia: FA Davis.
40. Cailliet, R. Knee pain and disability Philadelphia: FA Davis.
41. Cailliet, R. Hand pain and impairment Philadelphia: FA Davis.
42. Cailliet, R. Low back pain syndrome Philadelphia: FA Davis.
43. Cailliet, R. Soft tissue pain and disability Philadelphia: FA Davis.
44. Chaitow, L. Cranial manipulation: Theory and practice Edinburgh: Churchill Livingstone.
45. Greenman, P. E. Principles of manual medicine. Philadelphia: Lippincott Williams & Wilkins.
46. Wilson, A. Effective management of musculoskeletal injury: A clinical ergonomics approach to prevention. Churchill Livingstone.
47. O'Sullivan, F.A. Davis, Philadelphia. Physical rehabilitation: assessment and treatment.
48. Victor H. Frankel and Mangareta Nordin Basic Biomechanics of the Musculoskeletal system 2nd Edition
49. Essentials of Orthopedics for physiotherapists by John Ebenezer – Jaypee Publications
50. Practical Fracture Treatment by Ronald Mc Rae, Max Esser – Churchill Livingstone
51. Oxford Textbook of Orthopedics & Trauma – Christopher Bulstrode, Joseph Buckwalter, Oxford University Press
52. Fractures & Joint Injuries – By Watson Jones – Churchill Livingstone
53. Measurement in Physical Therapy – Churchill Livingstone, London

54. Soft Tissue Pain & Disability – Cailliet Rene, Jaypee Brothers, New Delhi

55. Physical therapy of the low back –Twomey, Churchill, Livingstone, London

56. Clinical Orthopaedic Examination by Ronald McRae – Churchill Livingstone

57. Campbell's operative orthopedics – By S. Terry Can ale, James H. Beaty – Mosby

58. Orthopedic Physical Assessment, By David J. Magee – Saunders

59. Diagnostic Imaging for Physical Therapists – by James Swain, Kenneth W. Bush & Juliette Brosing – Elsevier

60. Differential Diagnosis For Physical Therapists: Screening for Referral – by Catherine C. Goodman & Teresa Kelly Snyder – Saunders

61. Lynn Paul Taylor – Taylor's manual of physical evaluation and treatment

62. Goodman and Snyder – Differential diagnosis in physical therapy.

63. Leon Chaitow, and Judith Walker Delany - Clinical application on neuromuscular techniques: Vol-2 (The lower body).

1.20.12. **Specialty papers**

2) Master of Physiotherapy in NeuroSciences

1.20.12. A. SPECIALITY PAPER ONE

COURSE CODE-MPT-104

1. **MPT(N) 104: Clinical, Physical and Functional diagnosis in Neuro-Physiotherapy (NCPFD)**
2. **MPT (N) 202: Neurological Physiotherapy (NPT)**
3. **MPT (N) 203: Recent advances and Evidence Based Practice in Neuro-Physiotherapy (NRAEB)**

Course Title: MPT(N) 104: **Clinical, Physical and Functional diagnosis in Neuro-Physiotherapy** (NCPFD)

NCPFD 1.0. Course description

NCPFD 1.0.1. Course outcome

On successful completion of this unit, it is expected that students will be able to:

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in Neurology medicine & interpret clinical tests and special investigations commonly used in the diagnosis of these conditions.
2. Generate a primary physical diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on Neurological physiotherapy practice.
5. Demonstrate a broad range of technical skill in diagnosing the physiotherapy related neurology conditions.

SECTION- A

NCPFD 1.1. ICF conceptual frame work

NCPFD 1.2. Importance of assessment & evaluation, Outlines of principles and Methods of evaluation

NCPFD 1.3. Need and types of Documentation

NCPFD 1.4. Critical decision making and selection of outcome measures in Musculoskeletal Physiotherapy

NCPFD 1.5. Assessment, differential diagnosis and diagnosis of various Neurology conditions

NCPFD 1.6. Associated functional disturbances of higher function and their testing

NCPFD 1.7. Outcome measures used in Neuro-physiotherapy-for Cognitive impairment and disability, Focal disabilities, Global measures of disability, Motor impairment, ADL and extended ADL tests, Person with Disabilities and quality of life, Multiple Sclerosis, Parkinson ’s disease, Stroke, Head injury, Spinal cord injury, Pain scales

NCPFD 1.8. Clinical analysis of posture, movement and gait, use of gait analyzer

NCPFD 1.9. Principles, Techniques and interpretation of Pathological investigations and diagnostic imaging (CT, MRI, Ultra sound, PET, fMRI, bone scan and other diagnostic imaging) for diagnosis of neurological conditions.

NCPFD 1.10. Clinical examination and detection of movement dysfunction

NCPFD 1.11. Evaluation of ANS dysfunction with reference to Psycho physiological testing

NCPFD 1.12. Motor control assessment, reflexes and automatic reactions

NCPFD 1.13. Neurodevelopment assessment

SECTION- B

NCPFD 2.1. Assessment of Hand Function

NCPFD 2.2. Voluntary control assessment

NCPFD 2.3. Neuropsychological tests

NCPFD 2.4. Electrophysiological assessment devices – Instrumentation, Characteristics and components EMG (Qualitative and Quantitative EMG), NCV, Conventional Methods, RNS, EPS, EEG related to neurological disorders with interpretation.

NCPFD 2.5. Physical disability evaluation and disability diagnosis

NCPFD 2.6. Assessment of progressive locomotor disorder- Neuropathic, myopathic and NMJ conditions

NCPFD 2.7. Assessment and scales for diagnosis of pain

NCPFD 2.8. Biomarkers specific to neurological disorders

NCPFD 2.9. Assessment of Emotional Intelligence

NCPFD 2.10. Assessment of Peripheral nerve injuries and Cranial nerve disorders.

NCPFD 2.11. Neurophysiology and evaluation of Balance and Coordination

NCPFD 2.12. Assessment of Physical and Neurological Functions of Patients in ICU.

1.20.12.B. SPECIALITY PAPER TWO COURSE CODE: MPT (N)-202

MPT (N) 202: **Neurological Physiotherapy** (NPT)

NPT 1.0. Course Description

NPT 1.0.1. Course outcome

1. Develop a management plan, generally including some lifestyle factors, in cooperation with the Clinical Supervisor and consider a prognosis that reflects on the patient's problem.
2. Manage a patient in consultation and co-operation with the clinical supervisor, identifying the presenting problem, developing a basic working diagnosis and selecting a treatment regime that considers the presenting problem with consideration for ethical, practical and pragmatic concerns.
3. Maintain legal (accurate, clear and legible) patient histories, write basic referral letters and recognize the need of further referral in conference with Clinical Supervisor and peers.
4. Discuss the Common exercise prescriptions and their clinical use, and the sequence of treatment and how to advise different sorts of patients.

SECTION- A

NPT 1.1. History of neurological Physiotherapy, Epidemiology, classification of Neurology disorders, ICF classification of Neurological Disorders, symptomatology, patho-physiology and management of Neurological Disorders.

NPT 1.2. Physiotherapy interventions of various disorders of Central Motor control

NPT 1.3. Physiotherapy interventions of various disorders of the Motor Unit – Neuropathies, Myopathies and Neuromuscular Junction Disorders.

NPT 1.4. Physiotherapy interventions for Autonomic Nervous system dysfunction

NPT 1.5. Physiotherapy intervention for Peripheral Nervous system conditions (injuries and lesions)

NPT 1.6. Physiotherapy interventions for Tonal abnormalities.

NPT 1.7. Physiotherapy intervention for Traumatic conditions of CNS

NPT 1.8. Physiotherapy management for Demyelinating, Inflammatory, Infectious and Degenerative conditions.

NPT 1.9. Physiotherapy management for CNS Neoplasia.

NPT 1.10. Metabolic and Deficiency Disorders and their management

NPT 1.11. Congenital Neurological Disorders and management

NPT 1.12. Disorders of Perception & Cognition & their Rehabilitation,

NPT 1.13. Sensory System Dysfunction and rehabilitation

NPT 1.14. Oromotor Dysfunctions and Management

NPT 1.15. Visual Deficits and its management.

SECTION- B

NPT 2.1. Vestibular Dysfunction and its rehabilitation

NPT 2.2. Psychosomatic conditions and management.

NPT 2.3. Neuro - Surgical conditions and its postoperative management.

NPT 2.4. Neuro-Physiotherapy management in Intensive Care Units (ICU).

NPT 2.5. Physiotherapy interventions for muscle imbalances and corrective measures. Musculo-skeletal and Neurological complications of Locomotor Disorders

NPT 2.6. Pain Management

NPT 2.7. Adaptive and Assistive Functional Devices and technologies to improve neurological dysfunction.

NPT 2.8. Management of Bladder and Bowel Dysfunction

NPT 2.9. Neuro-physiotherapeutic approaches – Compensatory training approach, Muscle reeducation approach, Novel Approach, Neuro-physiological approaches - NDT, Brunnstrom, Roods, PNF, Sensory integration therapy. Motor relearning program, Constraint Induced movement therapy, Task Oriented approach, Novel approach, Vojta therapy**.** Biofeedback training, Neural mobilization and Neuro Dynamics, Sensory rehabilitation, Body Weight Supported Treadmill Training, Myofacial Release Technique, Inhibitory and Facilitation technique, Functional Re-Education, Learning skills, A.D.L, Tapping in neurological conditions.

NPT 2.10. FES, NMES, Biofeedback, Various equipment used in Neuro-physiotherapy

NPT 2.11. Problem Based Learning clinical conditions in Neurology physiotherapy.

NPT 2.12. Pharmacology in Neurophysiotherapy.

NPT 2.13. Training of Emotional Intelligence.

NPT 2.14. Hydrotherapy for Neurological conditions.

NPT 2.15. Palliative Care Approach.

NPT 2.16. Physiotherapy Management of Cerebellar Disorders.

1.20.12.C. SPECIALITY PAPER -THREE COURSE CODE: MPT (N)-203

MPT (N) 203: **Recent advances and Evidence Based Practice in Neuro-Physiotherapy** (NRAEB)

NRAEB 1.0.1. Course outcome

1. Understand and apply the information regarding recent advances in Neuro Physiotherapy for patient care.
2. Search the evidences available for assessment and management of neurological conditions.
3. Apply the evidences available for the management of various neurological conditions

SECTION- A

NRAEB 1.1. Basics of Genetic counseling, Stem cell therapy, Gene therapy

NRAEB 1.2. Recent advances in Pain Modulation and Rehabilitation.

NRAEB 1.3. Recent advances in Vocational Rehabilitation in Neurology Disorders with disability

NRAEB 1.4. Recent advancement in Neurology Orthosis – prescription and training.

NRAEB 1.5. Psychiatry problems in Neurological conditions and Physiotherapy (BAT, CBT). Psychological aspects of adaptation during various aspects of neurological disabilities

NRAEB 1.6. Institutional & community-based rehabilitation for Neurological Dysfunction**.**

NRAEB 1.7. Recent Neuro Physiotherapy technique - Mental Imagery technique, Virtual Reality Therapy/Virtual Clinic, Robotic Movement Therapy, Pilates therapy, Mirror Box therapy, Mime therapy, Floatation Therapy, Cupping Therapy, Jadestone Therapy, Matrix Rhythm Therapy, IASTM and Dry needling, CranioSacral therapy, Neurodynamics in Neurological conditions and Neural Mobilization, Hippo-therapy, Transcranial Direct Current Stimulation, Transcranial Magnetic Stimulation, Artificial Intelligence, Whole Body Vibrator and Neuromuscular Technique

NRAEB 1.8. Eclectic Approach.

SECTION- B

NRAEB 2.1. History of Evidence Based Practice in Neurological Physiotherapy, Clinical Decision Making, importance of Evidence Based Practice, Evidence about prognosis, experience and diagnosis, locating evidences, challenges and barriers in EBP.

NRAEB 2.2. Evidences in interventions for Neurological Impairments (Sensory, Motor, Cognitive and Perceptual)

NRAEB 2.3. Evidences for Physiotherapy in Traumatic CNS conditions

NRAEB 2.4. Evidences in Physiotherapy management of Stroke, Cerebellar Ataxia.

NRAEB 2.5. Evidences in Physiotherapy management of Peripheral Nerve Injuries

NRAEB 2.6. Evidences in Physiotherapy management of Parkinson's Disease

NRAEB 2.7. Evidences in Physiotherapy management of Myopathies, Neuropathies and NMJ Disorders

NRAEB 2.8. Sports training in Neurological Physiotherapy.

NRAEB 2.9. Tele rehabilitation in Neurological Physiotherapy

Books for Masters in Neurosciences

For paper III, IV, V.

1. American Psychological Association. Publication manual of the American Psychological Association. Washington, DC: Author.
2. Chichester, UK: John Wiley. Domholdt, E. Physical therapy research: Principles and applications, WB Saunders, Philadelphia, USA.
3. Kuzma, J. W., & Bohnenblust, S. E. Basic statistics for the health sciences. Boston: McGraw Hill.
4. Munro, B. H. Statistical methods for Healthcare research. Philadelphia: Lippincott.
5. Coakes, S. J., & Steed, L. G. SPSS: Analysis without anguish: Version 11.0 for Windows. Milton, Australia: John Wiley & Sons Inc. Jenkins, S., Price CJ, &Straker L.
6. The researching therapist. A practical guide to planning, performing and communicating research. Edinburgh: Churchill Livingstone.
7. Campbell, M.J., &Machin, D. Medical statistics: A commonsense approach. Chichester, UK: John Wiley.
8. Domholdt, E. Physical therapy research: Principles and applications. Philadelphia: WB Saunders.
9. Gowitzke,Williams and Wilkins.Scientific Basis of Human Movement .Baltimore..
10. Handbook of Physiology in Aging- Masoro, C.R.C. Press.
11. Hicks C:Research of Physiotherapists. Chrchhill Living stone, Edingburgh
12. Polgar S.: Introduction to Research in Health Sciences. Livingstone London.
13. Currier D.P: Elements of Research Physical Therapy. Williams & Wilkins, Baltimore.
14. Sproull: Hand Book of Research method. Scarecrow Press
15. Wilenski, Hale &Iremonger: Public Power and Administration.
16. Hickik Robert J: Physical Therapy Administration and management.
17. Nosse Lorry J: Management Principles for Physiotherapists.
18. Carpenter M.B: Human Neuroanatomy. Williams & Wilkins, Baltimore, n
19. Fraser: Physical Management of Multiple Handicapped. William & Wilkins, Baltimore

20. Aisen: Orthotics in neurological rehabilitation. Demos Publication, New York
21. Delisa: Manual of nerve conduction velocity techniques. Raven press, New York,
22. Kimura J, F.A Davis: Electrodiagnosis in diseases of nerve and muscle. Philadelphia ,
23. O' Sullivan, F. A Davis: Physical rehabilitation: Assessment and treatment. Philadelphia ,
24. Farber: Neuro – rehabilitation. W.B. Saimders , Philadelphia
25. Kerb D: Bio- Feedback – A practitioners guide. Guiford press.
26. Black I: The neural basis of motor control. Churchill, Livingstone, London -
27. Turnbull Gerode I: Physical therapy management of Parkinson's disease. Churchill, Livingstone, Londo -
28. Bobath B: Abnormal postural reflex activity caused by Brain Lesions. Aspen publications, Rockville
29. Eagel: Disord4e5r5s of Voluntary Muscle. Churchill, Living stone Edingburgh
30. Knot M. and Voss: Proprioception, neuro muscular facilitation techniques. Harper and Row, New York
31. Laidler, Capman and Hall: Stroke rehabilitation. London
32. Carr J.H, Shephered R.B: Motor relearning programme for stroke. Aspen publication, Rock Ville,
33. Bobath B. Heinmann: Adult hemiplegia evaluation and treatment: London
34. Brombley: Paraplegia and tetraplegia. Churchill, Livingstone, Edingburgh
35. Measurement in Physical therapy – Churchill, Livingstone, London
36. Maria stokes: Physical management neurological rehabilitation, Elsevier, Mosby.
37. Misra U.K, Kalita J: Clinical Neurophysiology NCV, EMG, Evoked Potentials, Elsevier, New Delhi,
38. Joel A Delisa, Gans B.M: Rehabilitation medicine principles and practice, revan, Philadelphia, New York,
39. Robert Gunzbng, MarekSzpalski: Whiplash Injuries, current concepts in prevention diagnosis and treatment, Lippincot Williams & wilkins.
40. Krusen's: Hand book of physical rehabilitation, kottke, lehmann, Saunder's Publications,
41. Ropper A.H, Brown R.H: Adam and victors principle of neurology, Mcgraw – hill companies USA
42. Richard S. Snell: Clinical Neuroanatomy for medical students, Lippincott Williams &wilkins
43. Martha Freeman Somers: Spinal cord injury functional rehabilitation
44. David S Butler: Mobilisation of the nervous system Churchill Livingstone, New York.
45. Darcy A. Umphred: Neurological rehabilitation, Mosby, Sydney,

46. Kenneth W. Lindsay, Ian Bone: Neurology & Neurosurgery illustrated,

47. M Flint Beal, Anthony.E. Lang, Albert Ludolph: Neurodegenerative Diseases, Cambridge University Publication, USA

48. Jose I. Suarez : Critical Care Neurology and Neurosurgery, HUMANA PRESS PUBLICATIONS,USA.

49. David R. Lynch : Neurogenetics-Scientific& Clinical Advances,Taylor& Francis Group Publication New York

50. Asbury, Mckann, Medonald: Diseases of Nervous System- Vol. I and Vol. II, Mcarthur public, 3rd edition.

1.20.13. **Specialty papers**

3) Master of Physiotherapy in Cardio- Pulmonary Sciences

MPT (C) 104: Clinical, Physical and Functional diagnosis in Cardio- Pulmonary Physiotherapy (CCPFD)

MPT (C) 202: Cardio- Pulmonary Physiotherapy (CPT)

MPT (C) 203: Recent advances and Evidence Based practice in Cardio- Pulmonary Physiotherapy (CRAEB)

1.20.13.A. SPECIALITY PAPER ONE COURSE CODE: MPT (C)-104

Course Title: Clinical, Physical and Functional diagnosis in Cardio- Pulmonary Physiotherapy (CCPFD)

CCPFD 1.0.1. Course outcome

1. Elicit and interpret clinical signs and symptoms of cardio-vascular and pulmonary diseases & interpret clinical tests and special investigations commonly used in the diagnosis of conditions.
2. Generate a primary diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images in various cardio-vascular and pulmonary disorders.

SECTION- A

CCPFD 1.1. ICF conceptual frame work

CCPFD 1.2. Importance of assessment & evaluation, Outlines of principles and Methods of evaluation Need and types of Documentation

CCPFD 1.3. Critical decision making and selection of outcome measures in cardiopulmonary Physiotherapy

CCPFD 1.4. GENERAL: Review of Anatomy, Embryology and Epidemiology of cardio-vascular, pulmonary and lymphatic pulmonary system.

CCPFD 1.5. Role of cardio respiratory therapist in patient assessment.

1. Patient clinician interaction and communication with assessment findings.
2. Confidentiality, concern and universal precautions.
3. A detailed and comprehensive cardio-respiratory health history.
4. Assessment standards, common scales, questionnaire indices used for patients with cardio-pulmonary dysfunction.

CCPFD 1.6. **Detailed assessment of cardio- vascular and pulmonary symptoms** (dyspnea, cough, sputum production, hemoptysis, clubbing, cyanosis, chest pain, syncope, fever, night sweating, headaches, altered sensorium, personality changes.

CCPFD 1.7. **Vital signs assessment**

1. Obtaining vital signs, clinical impressions
2. General clinical presentation
3. Temperature
4. Pulse including the peripheral pulses
5. Blood pressure
6. Respiratory rate

CCPFD 1.8. **Fundamentals of physical examination with diagnosis in cardiovascular and respiratory Physiotherapy**

1. Examination of head and neck
2. Lung topography – thoracic cage landmarks
3. Examination of Thorax/ pulmonary system
4. Examination of Precordium/cardiac system
5. Examination of Abdomen
6. Examination of Extremities

CCPFD 1.9. **Assessment of neonatal and pediatrics patients – new born, critically ill infants, older infants and child**

CCPFD 1.10. **Comprehensive geriatric assessment – age related sensory deficits, cardio-respiratory deficits and diagnostic tests, standard scales and questionnaires used in geriatric assessment**

CCPFD 1.11. **Nutritional assessment of patients with cardio- respiratory diseases**

CCPFD 1.12. **Fitness assessment**

1. Anthropometric and biophysical measurement and body composition
2. Flexibility tests and standards
3. Muscle strength and standard
4. Endurance tests and standards
5. Agility tests and coordination tests

CCPFD 1.13. **Exercise testing and standardization and interpretation**

1. TMT protocols- Maximal and submaximal protocols
2. Field protocols
3. Bicycle protocols
4. Step test protocols
5. Six minute walk test
6. Protocols for pediatric and geriatric population

CCPFD 1.14. **Interpretation and clinical relevance of investigations in cardio- pulmonary Physiotherapy**

1. Clinical laboratory studies – hematology, microbiology, urine analysis, histology, pathology
2. Pulmonary function tests – normal values
3. Spirometry, arterial blood gas analysis and its interpretation in cardio – respiratory Physiotherapy, capnography and pulse oximetry and its relevance in cardio- pulmonary Physiotherapy
4. Clinical application of chest radiograph – chest x-ray, examination, views; computed tomography, magnetic resonance imaging, lung scans - PET scan. Evaluation of chest radiography – clinical and radiographic findings in cardio-pulmonary disorders and its relevance cardio-pulmonary Physiotherapy
5. Laboratory and bedside interpretation of ECG findings – interpretation of normal and abnormal ECGs and its importance in cardio-respiratory physio-therapy and various ECG patterns in cardiac and lung disease
6. Cardio respiratory monitoring in critically ill patients including patients with artificial airways
    i. Ventilator assessment and evaluation of oxygenation in ICU
    ii. Assessment of cardiac output in ICU
    iii. Assessment of haemodynamic pressures in ICU
    iv. Clinical diagnosis in cardio- respiratory disorders in intensive care.

SECTION- B

CCPFD 2.1. Blood flow studies-arteriography, venography, Color Doppler, ANS testing and interpretation used in cardio- respiratory Physiotherapy and edema evaluation and interpretation.

CCPFD 2.2. Cardio respiratory assessment and diagnosis of patient on mechanical ventilator and interpretation of graphical forms, weaning modes and indices

CCPFD 2.3. Risk factor stratification, disability evaluation with reference to cardio vascular and pulmonary disorders

CCPFD 2.4. Psychological evaluation with reference to stress and anxiety in cardio- pulmonary disorders, Evaluation of stress and anxiety using various scales and questionnaires

CCPFD 2.5. Outcome measures used in Cardio – vascular and pulmonary Physiotherapy

CCPFD 2.6. Cardio-pulmonary Exercise Testing, VO2 max, METs – its importance in calculating energy expenditure and physical activities

CCPFD 2.7. Calculating energy expenditure using calorimetry method, various formulae and equations with emphasis on its importance in prescribing exercise in various patient population

CCPFD 2.8. Evaluation and diagnosis of sleep and breathing disorders.

1.20.13.B. SPECIALITY PAPER TWO

COURSE CODE: MPT (C)-202

Course Title: MPT (C) 202: Cardio and pulmonary Physiotherapy (CPT)

CPT 1.0.1. Course Outcomes:

1. Develop a management plan, generally including some lifestyle factors, incorporation with the Clinical Supervisor and consider a prognosis that reflects on the patient's problem.
2. Manage a patient in consultation and co-operation with the clinical supervisor, identifying the presenting problem, developing a basic working diagnosis and selecting a treatment regime that considers the presenting problem with some consideration for ethical, practical and pragmatic concerns.
3. Maintain legal (accurate, clear and legible) patient histories, write basic referral letters and recognize the need of further referral in conference with Clinical Supervisor and peers.
4. Discuss the Common exercise prescriptions and their clinical use, and the sequence of treatment and how to advise different sorts of patients.

SECTION- A

CPT 1.1. Principles of exercise prescription and exercise program adherence.

CPT 1.2. Components of physical fitness and Basic principles of exercise program design.

CPT 1.3. The art of science of exercise prescription in various patient population

CPT 1.4. Bioenergetics of exercise and training

CPT 1.5. Warm ups, stretching and cool down and its importance

CPT 1.6. Exercise program adherence and factors affecting exercise adherence.

CPT 1.7. Different forms of training methods.

CPT 1.8. Designing cardio-respiratory exercise programs for cardiac and pulmonary patients, geriatric and general population. Essentials of a C.R. exercise work- out, Aerobic training. Methods and modes, personalized programs.

CPT 1.9. Designing Resistance exercise programs.

1. Types of resistance training and developing respiratory exercise program including calisthenics.
2. Resistance exercise program for children and older adults.

CPT 1.10. Designing flexibility and stretching programs.

CPT 1.11. Designing weight management (weight loss and weight gain) and

CPT 1.12. Application of exercise prescription principles in various cardio-pulmonary disorders including edema management

SECTION- B

CPT 2.1. Nutrition and cardio-vascular and pulmonary diseases including diabetic population- Role of carbohydrates, proteins, fats, vitamins in health and disease.

CPT 2.2. Diet prescription in diabetic, hypertensive, cardio-metabolic syndromes, obesity and cancer patients according to calorie expenditure.

CPT 2.3. Exercise prescription/ physical activity in a high risk cardiac patient including L.V Dysfunction, chronic heart failure, myocardial ischemia.

CPT 2.4. Exercise prescription in prevention of CAD, obesity, renal dysfunction, diabetes mellitus, hypertension.

CPT 2.5. Cardio-vascular disorders and physiotherapy management including exercise prescription in:

i. Myocardial infarction

ii. Acquired heart conditions

iii. Hypertension, hypotension

iv. Rheumatic fever, rheumatic heart disease and non- rheumatic valvular diseases.

v. Diseases of myocardium, pericardial diseases, cardiomyopathies

vi. Vascular diseases, peripheral vascular diseases and lymphatic diseases

vii. Tumors of heart

viii. Athlete heart

ix. Congestive cardiac failure

x. Cardiac arrhythmias

xi. Congenital heart diseases

xii. Cardiac transplantation

SECTION- C

CPT 3.1. PULMONARY DISORDERS AND PHYSIOTHERAPY MANAGEMENT INCLUDING EXERCISE PRESCRIPTION IN:

i. Obstructive pulmonary diseases

ii. Restrictive pulmonary diseases

iii. Infective lung diseases

iv. Occupational lung diseases

v. Lung cancer

vi. Chest wall deformities and spinal cord injury

vii. Diaphragmatic diseases

viii. Sleep apnea/ hyperventilation syndrome

ix. Respiratory disorders in children, cystic fibrosis

x. COVID-19

CPT 3.2. Common emergency conditions in cardio-respiratory system in adults and children and ethical issues in intensive care

CPT 3.3. Management of Pediatric and geriatric Cardiac and pulmonary disorders

CPT 3.4. Burns rehabilitation in Critical Care unit

CPT 3.5. Cardio-pulmonary problems and complications in various neuromuscular disorders, facilitatory and inhibitory techniques and PNF techniques in various pulmonary disorders, manual techniques for various pulmonary disorders.

CPT 3.6. Physical agents used in various cardio-vascular and respiratory disorders

CPT 3.7. Cardio-vascular and pulmonary pharmacology- Indications, contraindications and effects and pharmacological management in cardiac and pulmonary disorders.

CPT 3.8. Body positioning: art and its physiological importance in general and in ICUs

CPT 3.9. Aerosol therapy- Principles and its role in physiotherapy.

CPT 3.10. Humidifiers and Atomizers role in physiotherapy.

CPT 3.11. Stress, Importance of exercise in stress management and various stress coping strategies, relaxation techniques including yogic postures and yogic breathing in various lifestyle disorders and other cardio-vascular and pulmonary conditions.

CPT 3.12. Importance of Patient education and counseling in various cardio-vascular and pulmonary disorders in cardio- respiratory conditions, CBR in cardio vascular and respiratory conditions.

CPT 3.13. Role of Tele-rehabilitation in cardiac and pulmonary disorders.

CPT 3.14. Clinical decision making in Cardiovascular and pulmonary physiotherapy.

1.20.13.C. SPECIALITY PAPER THREE

COURSE CODE: MPT (C)-203

MPT (C) 203: Recent advances and Evidence Based Practice in Cardio and pulmonary Physiotherapy (CRAEB)

CRAEB 1.0.1. Course Outcome

1. Understand and apply the information regarding recent advances in cardio-pulmonary physiotherapy for patient care.
2. Search the evidences available for assessment and management of cardiopulmonary conditions.
3. Apply the evidences available for the management of various cardio-pulmonary conditions

SECTION- A

CRAEB 1.1. GENERAL:

i. Optimizing treatment prescription: relating treatment to the underlying pathophysiology of cardio-vascular and pulmonary disorders- an evidence-based practice

ii. Documentation of the data, Report writing –prescription of exercises

iii. Importance of creating awareness in community, Patient education and psychological counseling in various cardio-vascular and pulmonary disorders evidence-based practice

iv. Recent advancement in Cardio- pulmonary resuscitation (basic and advanced)

CRAEB 1.2. Bronchial hygiene- Physiological basis and clinical application, evidence-based practice and recent advances of airway clearance techniques, including Facilitating airway clearance with coughing techniques.

CRAEB 1.3. Care of a dying patient. – Ethical issues and recent guidelines

CRAEB 1.4. Cardiopulmonary training in various patient populations. Athletes, Geriatric and pediatric population

CRAEB 1.5. Medical gas therapy including oxygen therapy**:** physiological basis, modes of administration, and home delivery care- an evidence-based practice and recent advances including hyperbaric oxygen therapy.

CRAEB 1.6. Aerosol therapy**-** An Evidence based practice in chest physiotherapy.

SECTION- B

CRAEB 2.1. Recent advances and evidence-based practice in Exercise testing, planning, principles of exercise prescription and PT management in cardio- vascular and pulmonary conditions.

CRAEB 2.2. Recent advances and evidence base practice in cardio-respiratory Physiotherapy and exercise prescription in special populations like cancer, renal conditions, burns, abdominal surgeries, Neurological patients and Diabetic mellitus patients.

CRAEB 2.3. Recent advances in the use of physical agents and PT management in wounds, ulcers, grafts and incisions and vascular disorders.

CRAEB 2.4. Evidence based practice of core muscle strengthening, resistance training, endurance training, and other training methods in cardiac and pulmonary rehabilitation

CRAEB 2.5. Pilates- school of thought for cardiopulmonary conditions.

CRAEB 2.6. Physiotherapy management in oncology- Evidence based practice and recent advances.

CRAEB 2.7. Recent advances and evidence-based practice in Respiratory Physiotherapy training techniques and respiratory Physiotherapy devices.

CRAEB 2.8. Evidence based practice and recent advances in improving Cardio-respiratory fitness training in all populations including general, pediatric and geriatric population.

CRAEB 2.9. Evidence based practice and Recent guidelines in cardiac rehabilitation and pulmonary rehabilitation

CRAEB 2.10. Role of exercise and quality of life and cardio-pulmonary rehabilitation, health status measurements and recent advances

CRAEB 2.11. Use of advance Assistive devices like Robot therapy, continuous lateral rotation therapy, intrapulmonary percussive ventilator and technologies in Cardiovascular and pulmonary system.

CRAEB 2.12. Evidence based practice and recent advances of Aquatic therapy in Cardiovascular conditions like diabetes, PVD, hypertension etc.

BOOKS for Physiotherapy in Cardio Pulmonary Sciences:

1) Froelicher /Myers- “Exercise and heart’ Saunders publication.
2) Jean Jobin et al. Advances in Cardio-Pulmonary Rehabilitation”
3) Scot Irvin, Lan Stephen Tecklin- “Cardio-Pulmonary physical therapy-a guide to practice”, Mosby.
4) Frances J Brannon, Margaret W Foley, Julie Ann Stars, Lauren M Saul
5) “Cardio-Pulmonary Rehabilitation-Basic Theory and Application”, F A Davis Company.
6) Cynthia Coffin Zadai- “Pulmonary management in Physical therapy”, Churchill Livingstone.
7) Barbara A Webber and Jennifer A Pryor- “Physiotherapy for respiratory and cardiac problems”, Churchill Livingstone.
8) George G. Burton, John E Hodgin, Jeffrey J Ward- “Respiratory Care-A Guide to Clinical Practice” 4th edition, Lippincott Williams and Wilkins,
9) Robert MBerne, Matthew N Levy- “Cardio-vascular physiology”, Mosby.
10) John B. West- “Respiratory Physiology-the essentials”, Lippincott Williams and Wilkins.
11) Macleod’s Clinical Examination.
12) Andrews Davies and Carl Moores- “The Respiratory System”, illustrated by Robert Britton, Churchill Livingstone.
13) George G. Burton, John E Hodgin, Jeffrey J Ward- “Respiratory Care-A Guide to Clinical Practice”, Lippincott Williams and Wilkins,
14) Richard d Branson/Robert L Chatburn- “Respiratory Care Equipment”, J B Lippincott Company.
15) N R Malentyre/R D Branson- “Mechanical Ventilation”, Saunders.
16) Joanne Watchie- “Cardio-Pulmonary Physical Therapy”, Saunders.
17) Hillegass and Sadowsky. “Essentials of Cardio-Pulmonary Physical Therapy”, Saunders, Elsevier.
18) Michael L. Pollock and Donald H Schmidt- “Heart disease and Rehabilitation”.
19) Scot Irvin, Lan Stiphen Tecklin. “Cardio-Pulmonary physical therapy-a guide to practice”, Mosby.
20) Frances J Brannon, Margaret W Foley, Julie Ann Stars, Lauren M Saul
21) Cardio-Pulmonary Rehabilitation-Basic Theory and Application”. F A Davis Company

1.20.14 Specialty papers

4) Master of Physiotherapy in Sports Sciences

MPT (S) 104: Sports traumatology (STT)

MPT (S) 202: Concepts in sports medicine (SSM)

MPT (S) 203: Recent advances and Evidence Based practice in Sports Physiotherapy (SRAEB)

1.20.14.A. SPECIALITY PAPER - ONE COURSE CODE: MPT (S)-1O4

Course Title: MPT (S) 104: **SPORTS TRAUMATOLOGY (STT)**

SECTION- A

STT 1.1. ICF conceptual frame work

1. Importance of assessment & evaluation, Outlines of principles and Methods of evaluation Need and types of Documentation
2. Critical decision making and selection of outcome measures in SPORTS Physiotherapy
3. Investigative Procedures**.** Diagnostic imaging (CT, MRI, Ultra sound, bone scan and other diagnostic imaging's) for diagnosis of congenital anomalies and normal variants, traumatic injuries, scoliosis, degenerative disorders and infections)
4. Principles of pathological investigations and imaging techniques related to musculoskeletal disorders with interpretation Causes & Mechanism of Sports Injuries

STT 1.2. Evaluaton Of Risk Factors And Pre-Participation Examination:

1. Components of pre-participation evaluation, Scope and implementation of pre-participation program in Sports PT
2. Evaluation of Physical Fitness: Assessment of components of physical fitness including functional tests: muscle strength, flexibility, agility, balance, co-ordination, sensory deficits, cardio-pulmonary endurance
3. Sports-Specific evaluation and criteria for return to sport
4. Examination of lower limb
    i. Pelvis
    ii. Hip
    iii. Thigh
    iv. Knee
    v. Leg
    vi. Ankle and Foot

5. Examination of Upper Extremity

   i. Shoulder girdle

   ii. Shoulder

   iii. Arm

   iv. Elbow &Forearm

   v. Wrist and hand.

6. Assessment of vertebral column:

   i. Cervical

   ii. Thoracic

   iii. Lumbosacral including Tests of Neural Tension

7. Sporting emergencies screening

   i. Head and neck

   ii. Face

   iii. Abdominal injuries

8. Anthropometric evaluation

9. Kinesiological EMG

SECTION- B

STT 2.1. Causes & Mechanism of Sports Injuries

STT 2.2. Prevention of Sports injuries

STT 2.3. Principle of management of sports injuries

STT 2.4. Common acute and overuse injuries, causation, prevention and management of lower Extremity in Sports PT

STT 2.5. Common acute and overuse injuries, causation, prevention and management of upper Extremity in Sports PT

STT 2.6. Common sports injuries of spine with respect to causation, prevention and management

STT 2.7. Sporting emergencies first aid

STT 2.8. Emergency Medical Planning and Cover for Sports Events

STT 2.9. Emergency Situations, Primary and secondary emergency assessment, emergency plan, Transportation of an injured person

STT 2.10. Treatment of collapsed athlete- Severe head injury, Athlete with spinal injury,

STT 2.11. Causes of Collapse

STT 2.12. Sports specific injuries, with special emphasis on the specific risk factor, nature of Sports,

STT 2.13. Biomechanical Analysis of Skills, kind of medical intervention anticipated and prevention with respect to various sporting events

i. Individual events: Field & Track

ii. Team events

iii. Contact and Non-contact sports

iv. Water sports

1.20.14.B. SPECIALITY PAPER –TWO

COURSE CODE: MPT (S)-202

Course Title: MPT (S) 202: Concepts in sports medicine (SSM)

SECTION- A

Sports Training Parameters and Methods

SSM 1.1. Training Load, Adaptation and Recovery: Relationship of load and recovery, physiotherapeutic and psychological means of Recovery, Variables of Training:

Volume, Intensity, Density, Complexity.

SSM 1.2. Relationship between volume and intensity

SSM 1.3. Fatigue and overtraining: Diagnosis, Monitoring and preventing overtraining. RECOVERY METHODS

SSM 1.4. Training Methods: Interval training, Continuous training, Circuit training, Fartlek training, Weight training, Plyometric method, Cross training

SSM 1.5. Bio Motor Abilities and Program Design

SSM 1.6. Anaerobic Exercise Training & Prescription: Prerequisites, types and Factors affecting the training variables: Strength Development, Plyometric Training, Speed, Agility and Speed Endurance Development

SSM 1.7. Aerobic Exercise Training & Prescription: Prerequisites, types and Factors affecting the training variables

SSM 1.8. Coordination Training**:** Definition, Classification of coordinative abilities, factors affecting coordination and Methods to develop coordination

SSM 1.9. **Sports Psychology-**Role of Sports Psychology in Sports performance, Factors affecting growth and development & role of heredity & environment Biofeedback, Mental coping strategies, Visual Imagery, Meditation History and current status of Sports Psychology

i. Personality assessment and sports personality · Attention and perception in sports

ii. Concentration training in sports · Motivational orientation in sports

iii. Pre-competitive anxiety · Relaxation training · Aggression in sports · Role of Psychology in dealing with injuries · Eating disorders · Goal setting (Psychological aspect of doping, stress management, group behaviour and leadership, emotion

SSM 1.10. Para SPORTS

SSM 1.11. Sports Massage

SECTION- B

Non traumatic conditions of athletes

SSM 2.1. General Illness

SSM 2.2. Chronic/ non-communicable diseases

SSM 2.3. Exercise Induced Asthma

SSM 2.4. Anemia

SSM 2.5. Delayed onset muscle soreness (DOMS)

SSM 2.6. Runner's high & Exercise addiction

SSM 2.7. G.I.T. Diseases

SSM 2.8. Eating disorders in athletes

SSM 2.9. AIDS in sports people

SSM 2.10. **Sports for diseased**

1. Exercises and congestive heart failure
2. Exercise for Post coronary & bye pass patients
3. Exercise for diabetics
4. Diagnosis and management of skin conditions of Athletes:
    a. Bacterial infections
    b. Fungal Infections
    c. Viral infections
    d. Boils
    e. Cellulites.

SSM 2.11. **Female athlete problems**

i. Sports Amenorrhea.

ii. Injury to female reproductive tract.

iii. Menstrual Synchrony.

iv. Sex determination.

v. Exercise and pregnancy

1.20.14.C. SPECIALITY PAPER -THREE

COURSE CODE: MPT (S)-203

Course Title: Recent advances and evidence-based practice in Sports Physiotherapy (SRAEB)

SECTION A

SRAEB 1.1. Exercise and Common Pulmonary Conditions

i. Exercise induced bronchial obstruction

ii. Exercise in chronic airway obstruction

iii. Air pollution and exercise

SRAEB 1.2. Exercise and Cardiac Conditions

i. Exercise prescription for heart disease

ii. Exercise in primary prevention in ischemic heart disease

iii. Exercise for secondary prevention of ischemic heart disease

SRAEB 1.3. Diabetes and Exercise

i. Exercise in diabetic patients

ii. Exercise as a method of control of diabetes

SRAEB 1.4. Protective equipment design of shoe safety factors in equipment. Health club and fitness concept, use and misuse of equipment

SRAEB 1.5. Special concerns for para-athletes

SECTION B

SRAEB 2.1. Exercises for special categories

i. Child and adolescent athlete's problems (Exercise for growing bones)

ii. Special problems of older athletes

iii. Sports and exercise programme for geriatrics and rheumatic population

SRAEB 2.2. Doping in Sports

i. IOC prohibited drugs- groups and classifications

ii. IOC rules and regulations on doping in sports hazards of prohibited substances

SRAEB 2.3. Identification of talent for sports –

i. Detailed procedure for screening and identification of sports talent

ii. Prediction of adult potentials at the young age.

SRAEB 2.4. Sports Pharmacology and medico-legal issues in sports

SRAEB 2.5. Segmental Stabilization Concepts of spine

SRAEB 2.6. Emergency medical planning and cover for Sports events

SRAEB 2.7. Effect of physical activity intervention in youth

SRAEB 2.8. Precision heart rate training

SRAEB 2.9. Current concepts in obesity management XIII· Electromyography and Rehabilitation

SRAEB 2.10. Current concepts in comprehensive physical examination for the instabilities of knee

SRAEB 2.11. Current concepts in tendinopathies

BOOKS for Physiotherapy in Sports Medicine

1. Chew, F. (110107). Skeletal radiology: The bare bones (2nd ed.). Baltimore, MD: Williams & Wilkins.
2. Eisenberg, R. L., & Johnson, N. M. (2003). Comprehensive radiographic pathology (3rd ed.). St Louis, MO: Mosby.
3. Hughes, J., & Hughes, M. (110107). Imaging: Picture tests. Edinburgh: Churchill Livingstone.
4. Mace, J. D., &Kowalczyk, N. (110104). Radiographic pathology for technologists (2nd ed.). St Louis, MO: Mosby.
5. Redhead, D. N. (110105). Imaging: Colour guide. Edinburgh: Churchill Livingstone.
6. Yochum, T. R., & Rowe, L. R. (2005). Yochum and Rowe's essentials of skeletal radiology (3rd ed., Vols. 1-2). Baltimore, MD: Lippincott Williams & Wilkins.
7. Nolte, J., & Angevine, Jr. J. B. (2000). The human brain in photographs and diagrams (2nd ed.). St Louis, MO: Mosby.
8. Wicke, L. (110107). Atlas of radiologic anatomy (6th ed.). Munich, Germany: Lea &Febiger.
9. Seidel, H. (110105). Mosby's guide to physical examination. St Louis, MO: C.V. Mosby.
10. Cailliet, R. Neck and arm pain Philadelphia: FA Davis.
11. Cailliet, R. Shoulder pain Philadelphia: FA Davis.
12. Cailliet, R. Knee pain and disability Philadelphia: FA Davis.
13. Cailliet, R. Hand pain and impairment Philadelphia: FA Davis.
14. Cailliet, R. Low back pain syndrome Philadelphia: FA Davis.
15. Cailliet, R. Soft tissue pain and disability Philadelphia: FA Davis
16. O'Sullivan, F.A. Davis, Philadelphia 110104. Physical rehabilitation: assessment and treatment.

17. Kuprian: Physical Therapy for Sports, W.B. Saunders
18. Malone: Orthopeadic and Sports Physical Therapy, C.V. Mosby.
19. Zulunga et al: Sports Physiotherapy, W.B. Saunders.
20. Reed: Sports Injuries – Assessment and Rehabilitation, W.B. Saunders.
21. Gould: Orthopaedic Sports Physical Therapy, Mosby.
22. Norris: Sports Injuries – Diagnosis and Management for Physiotherapists, Heinmann.
23. Gait analysis – Perry J., Black Thorofare, New Jersy, 110102.
24. Mc Ardle, Katch, Katch: Exercise Physiology Edition IV.
25. Era Volinski: Nutrition and exercise in Sports - CRC Press, New York.
26. George A. Brooks, Thomas D. Fahey: Exercise Physiology – Human Bioenergetics and its applications 11084, John Wiley & Sons, New York.
27. Astrand&Rodahl: Text Book of Work Physiology, McGraw Hill.
28. Fox and Mathews - The Physiological Basis of Physical Education and athletics – Holt Saunders.
29. Erston and Reilly - Kinanthropometry and Exercise Physiology Laboratory Manual tests, Procedures and Data - F & FN Spon Madras.
30. Rowland - D4e7v4elopmental Exercise Physiology - Human Kinetics.
31. Clarke - Exercise Physiology - Prentice Hall.
32. Gardiner M. Dena: The Principles of Exercise Therapy - CBS Publishers Delhi.
33. Kisner and Colby: Therapeutic Exercises – Foundations and Techniques, F.A. Davis.
34. Basmajian John V.: Therapeutic Exercise, Williams & Wilkins.
35. Wood & Baker: Beard's Massage, W.B. Saunders.
36. William E. Prentice: Rehabilitation Techniques - Mosby.
37. Werner Kuprian: Physical Therapy for Sports, W.B. Saunders.
38. Kennedy: Mosby's Sports Therapy Taping Guide.
39. Malone: Orthopeadic and Sports Physical Therapy, C.V. Mosby.
40. William E. Prentice: Therapeutic Modalities in Sports Medicine - Mosby.
41. William E. Prentice: Rehabilitation Techniques - Mosby.
42. O' Sullivan, Schmitz: Physical Rehabilitation – Assessment and Treatment - F.A. Davis.
43. John Low & Reed: Electrotherapy Explained, Butterworth.
44. Meryl Roth Gersh: Electrotherapy in Rehabilitation, FA Davis.

45. Joseph Kahn: Principles and Practice of Electrotherapy, Churchill Livingstone.

46. Harrelson and Andrews: Physical Rehabilitation of Injured Athlete.

47. Nelson and Currier: Clinical Electrotherapy, Prentice Hall.

48. Greenman: Principles of Manual medicine, William and Wilkins.

49. Kuprian: Physical Therapy for Sports, W.B. Saunders.

50. Bates: Aquatic Exercise Therapy, W.B. Saunders.

51. Michlovitz - Thermal agents in Rehabilitation - F.A. Davis.

52. Lehmann - Therapeutic Heat and Cold - Williams & Wilkins

53. Morgan and King: Introduction to Psychology - Tata McGraw Hill.

54. Suinn: Psychology in Sports: Methods and applications, Surjeet Publications.

55. Grafiti: Psychology in contemporary sports, Prentice Hall.

56. Manual of nerve conduction velocity techniques – De Lisa, Raven press, New York, 11082.

57. Physical rehabilitation: assessment and treatment – O'Sullivan, F.A. Davis, Philadelphia 110104.

58. Bio-feedback – A practitioners guide – Kerb D, Guiford press

59. James G. Hay – The Biomechanics of Sports Techniques, Prentice Hall. Brunnstrom - Clinical Kinesiology, F.A. Davis.

60. Luttgens K., Hamilton N.: Kinesiology – Scientific Basis of Human Motion, Brown & Benchmark.

61. Kreighbaum E., Barthels K.: Biomechanics – A Qualitative approach for studying Human Motion, MacMillan.

62. Rasch and Burk: Kinesiology and Applied Anatomy, Lee and Fabiger.

63. White and Punjabi - Biomechanics of Spine - Lippincott.

64. Norkin&Levangie: Joint Structure and Function - A Comprehensive Analysis - F.A. Davis.

65. Kapandji: Ph47y5siology of Joints Vol. I, II & III, W.B. Saunders.

66. Northrip et al: Analysis of Sports Motion: Anatomic and Biomechanic perspectives, W.C. Brown Co., IOWA.

67. Leveac B.F.: Basic Biomechanics in Sports and Orthopeadic Therapy, C.V. Mosby.

68. Morris B. Mellion: Office Sports Medicine, Hanley & Belfus.

69. Richard B. Birrer: Sports Medicine for the primary care Physician, CRC Press.

70. Torg, Welsh &Shephard: Current Therapy in Sports Medicine III - Mosby.

71. Zulunga et al: Sports Physiotherapy, W.B. Saunders.

72. Brukner and Khan: Clinical Sports Medicine, McGraw Hill.

73. Reed: Sports Injuries – Assessment and Rehabilitation, W.B. Saunders.

74. Gould: Orthopaedic Sports Physical Therapy, Mosby.

75. C. Norris: Sports Injuries – Diagnosis and Management for Physiotherapists, Heinmann.

76. D. Kulund: The Injured Athlete, Lippincott.

77. Nicholas Hershman:

78. Vol. I The Upper Extremity in Sports Medicine.

79. Vol. II The Lower Extremity and Spine in Sports Medicine.

80. Vol. III The Lower Extremity and Spine in Sports Medicine. Mosby.

81. Lee & Dress: Orthopaedic Sports Medicine - W.B Saunders.

82. Fu and Stone: Sports Injuries: Mechanism, Prevention and Treatment, Williams and Wilkins.

83. Scuderi, McCann, Bruno: Sports Medicine – Principles of Primary Care, Mosby.

84. First Aid to Injured: St. John's Ambulance Association.

85. Andrea Bates and Norm Hanson: Aquatic Exercise Therapy, W.B. Saunders.

86. Dvir: Isokinetics: Muscle Testing, Interpretation and Clinical Applications,

87. W.B. Saunders.

88. Hartley: Practical Joint Assessment, A Sports Medicine Manual, upper and lower quadrants, C.V. Mosby.

89. Albert: Eccentric Muscle Training in Sports and Orthopedics, W.B. Saunders.

90. Voss et al - Proprioceptive Neuromuscular Facilitation - Patterns & Techniques - Williams & Wilkins

91. Torg, Welsh andShephard: Current Therapy in Sports Medicine III - Mosby.

92. Reed: Sports Injuries – Assessment and Rehabilitation, W.B. Saunders.

93. Nordinand Frankel: Basic Biomechanics of Muscular Skeletal System: Williams and Wilkins.

94. Mc Ardle, Katch, Katch: Exercise Physiology.

95. Brukner and Khan: Clinical Sports Medicine, McGraw Hill.

96. O'Leary: Drugs and Doping in sports.

97. Wilson, A. Effective management of musculoskeletal injury: A clinical ergonomics approach to prevention. Churchill Livingstone.

98. Lee and Dres4s7:6 Orthopaedic Sports Medicine - W.B Saunders

99. Kurt Dorr and Jonathan S. Rakich: *Hospital Organization and Management:*

1.20.15 Specialty papers

5) Master of Physiotherapy in Pediatrics and Neonatal Sciences:

MPT (P) 104: Clinical, physical & functional diagnosis in pediatric physiotherapy (PCPFD)

MPT (P) 202: Pediatric physiotherapy (PPT)

MPT (P) 203: Recent advances and Evidence Based practice in PEDIATRIC PHYSIOTHERAPY (PRAEB)

SPECIALITY PAPER -ONE

1.20.15.A. COURSE CODE: MPT (P)-104

Course Code: CLINICAL, PHYSICAL & FUNCTIONAL DIAGNOSIS IN PEDIATRIC & NEONATAL PHYSIOTHERAPY MPT (P) 104 (PCPFD)

PCPFD **1.0.1. Course outcome**

On successful completion of this unit, it is expected that students will be able to:

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in Pediatric (neurology, cardio-respiratory, musculoskeletal) medicine & in- terpret clinical tests and special investigations commonly used in the diagnosis of conditions.
2. Generate a primary diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Explain the medical management of various conditions typically presented in Pediatric disorders.
5. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on Pediatric physical therapy practice.
6. Demonstrate a broad range of technical skills, including the ability to manage common pediatric conditions.

SECTION- A

PCPFD 1.1. Review of Embryology

PCPFD 1.2. Maturational, patho -physiological & recovery process in the CNS

PCPFD 1.3. Genetic basis of pediatrics disorders

PCPFD 1.4. Pain assessment in neonates & children

PCPFD 1.5. Patho-mechanics and clinical biomechanics of posture and movement in various Pediatric conditions

PCPFD 1.6. Analysis and diagnosis of functional mechanics and patho-mechanics of gait in children

PCPFD 1.7. Principles, procedure, interpretation and significance of Diagnostic imaging (CT, MRI, Ultra sound, bone scan, PET scan, fMRI) for clinical and functional diagnosis in various orthopedic, cardio-respiratory and neurological conditions in children

PCPFD 1.8. Clinical examination in general and physical and functional diagnosis for detection of movement dysfunction

PCPFD 1.9. Principles of pathological investigations, Electro-diagnosis and its interpretation related to common pediatric disorders- Laboratory investigation, clinical tests (EEG, ECG, Evoked potentials, qualitative and quantitative EMG, NCV & Biofeedback)

PCPFD 1.10. Evaluation of typical and atypical development of children in various domains of development (Gross, fine, cognitive, speech & language, personal social and adaptive functions)

PCPFD 1.11. Evaluation, epidemiology, symptomatology and patho-physiology of common Pediatric congenital, cardio-respiratory, neurological and musculo-skeletal disorders

PCPFD 1.12. Clinical, physical and functional diagnosis of developmental disorders

PCPFD 1.13. Neurodevelopment assessment

PCPFD 1.14. Hand Function-Assessment and diagnosis

PCPFD 1.15. Theories of Motor control and Motor learning processes

PCPFD 1.16. Principles, administration and interpretation of Developmental screening tools

SECTION- B

PCPFD 2.1. Voluntary control assessment

PCPFD 2.2. Outcome measures used in Pediatric Physiotherapy

PCPFD 2.3. Pre and post- surgical physiotherapeutic (Physical and functional) evaluation for various surgical conditions in children

PCPFD 2.4. Anthropometrics measurements in children- Principles, methods, normal values for different ages, deviation and its clinical and functional significance

PCPFD 2.5. Exercise testing & Physical fitness assessment in children with & without disability (Range of motion, Muscle strength, endurance and skills, Body composition, Cardiac efficiency tests and spirometry

PCPFD 2.6. Fitness evaluation in children for sports

PCPFD 2.7. Physical and functional assessment for Aids, appliances& adaptive devices in Pediatric disorders

PCPFD 2.8. Physical disability evaluation and disability diagnosis

PCPFD 2.9. Assessment of various pediatric medical and surgical conditions

1.20.15.B. SPECIALITY PAPER -TWO COURSE CODE: MPT(P) -202

MPT (P) 202: Pediatric & Neonatal physiotherapy (PPT)

PPT 1.0. Course outcomes

1. Demonstrate an understanding of dysfunctions affecting Pediatric musculoskeletal, neurological and cardio -respiratory system including their patho- physiology.
2. Demonstrate a range of technical skills related to Pediatric therapy such as NDT, Sensory integration concept, classification and their application following diagnosis of dysfunction, indication, contraindication and adjunct therapies.
3. Demonstrate specific rehabilitation skills, principles of rehabilitation of Pediatric disorders.
4. Explain factors involved in effective management of patients and also justify the importance of preventive care in rehabilitation

SECTION- A

PPT 1.1. Genetic counseling

PPT 1.2. Physiotherapy management of growth and developmental disorders (gross motor, fine, speech & language, personal- social –adaptive

PPT 1.3. Therapeutic techniques used in Neuro- pediatric conditions- Handling & positioning techniques, NDT, Vojta, Roods, CIMT, Sensor-motor re-education, PNF, Peto, Temple Fay, Phelps

PPT 1.4. Adjunct therapies- Manipulation, mobilization, taping, MFR, Cranio-sacral therapy, Body suits, hydrotherapy, hippo-therapy

PPT 1.5. Pain control & management in children

PPT 1.6. Motor learning techniques

PPT 1.7. Sensory integration disorders and management

PPT 1.8. Management of perceptual and cognitive disorders

PPT 1.9. Play behavior & its clinical application in therapy

PPT 1.10. Integrated approach in management of Pediatric disorders

PPT 1.11. Neonatal care and early intervention for risk babies

PPT 1.12. Physiotherapy management for congenital loco-motor disorders including prosthetic and orthotic prescription

PPT 1.13. Pediatric disability management at institutional & community levels

PPT 1.14. Pre and Post-operative management of pediatric surgeries

PPT 1.15. Rehabilitation of common pediatric musculo-skeletal disorders

PPT 1.16. Management of progressive loco-motor disorders- Neuropathic and Myopathic conditions

SECTION- B

PPT 2.1. Management of learning disabilities, ADHD, Autism, Developmental coordination disorders and behavioral disorders

PPT 2.2. Physiotherapeutic management of A.D.L and functional activities

PPT 2.3. Sports training in pediatrics

PPT 2.4. Psychological and mental health problems in children

PPT 2.5. Management of Child abuse and its associated problems

PPT 2.6. Management of common congenital, neurological, musculo-skeletal and cardio-respiratory disorders

PPT 2.7. Vocational rehabilitation for pediatric disorders

PPT 2.8. Metabolic disorders and their management

PPT 2.9. Exercise prescription for pediatric disorders

PPT 2.10. Oromotor dysfunction in children

1.20.15.C SPECIALITY PAPER THREE COURSE CODE: MPT (P)-203

MPT (P) 203: Recent advances and Evidence Based practice in PEDIATRIC &NEONATAL PHYSIOTHERAPY (PRAEB)

PRAEB 1.0.1. Course outcomes

1. Understand and apply the information regarding recent advances in Pediatric Physiotherapy for patient care.
2. Search the evidences available for assessment and management of Pediatric conditions.
3. Apply the evidences available for the management of various Pediatric conditions

SECTION- A

PRAEB 1.1. Advanced instruction in physical examination, diagnosis, treatment and reassessment of the Pediatric neurological, musculoskeletal, cardio – respiratory system

PRAEB 1.2. Psychosocial affects in children and parents

PRAEB 1.3. Evidence based practice for exercise prescription for home program

PRAEB 1.4. Report writing for clinical cases & research

PRAEB 1.5. Recent advances in prescription, indications, assessment and training for orthosis, prosthesis and adaptive equipment in physically challenged children

PRAEB 1.6. EBP in Muscul4o81skeletal and Neurological loco-motor disorders in children

SECTION- B

PRAEB 2.1. Rationale of basic and advanced investigative procedures with differential diagnosis

PRAEB 2.2. EBP & recent advances on the role of Physical therapy in public and special schools-

PRAEB 2.3. Recent advances in exercise prescription for children

PRAEB 2.4. EBP for management of pediatric oncology & burns

PRAEB 2.5. Recent advances in Pain control, assessment & management in children

PRAEB 2.6. Equipment's, assessment & treatment in neonatal & pediatric intensive care units

PRAEB 2.7. Recent advances in instrumentations, theories, handling and pediatric physical therapy techniques

PRAEB 2.8. Problem based learning relevant to clinical conditions typically seen in pediatrics

Books for Pediatrics and neonatal sciences

1. Scientific basis of human movement –Gowitzke, Williams and Wilkins, Baltimore,
2. Clinical biomechanics of spine – White A, and Panjabi- J, B. Lippincot, Philadelphia
3. Human Neuroanatomy – Carpenter M.B. Williams & Wilkins, Baltimore,
4. Physical therapy in early infancy – Wilhelm, Churchill Livingstone, New York
5. Physical therapy for children – Campbell Suzann K. W.B Saunders, Philadelphia,
6. Physical management of multiple handicapped – Fraser, William and Wilkins, Baltimore.
7. Elements of paediatric Physiotherapy – Eckersley, Churchill Livingstone, Edinburgh,
8. Physiotherapy in paediatrics - Shepherd R Heinmann, London,
9. The growth chart – WHO, Geneva,
10. Orthotics in neurological rehabilitation – Aisen, Demos Publication, New York
11. Electrodiagnosis in diseases of nerve and muscle – Kimura J, F.A. Davis, Philadelphia.
12. Orthopaedic physical therapy – Donatteli, London, Churchill Livingstone,
13. Gait analysis – Perry J., Black Thosofare, New Jersey,
14. Biofeedback – A practitioner's guide – Kerb D, Guilford press.
15. Abnormal postural reflex activity caused by Brain lesions – Bobath B. Aspen publications, Rockville, 1897.
16. Disorders of voluntary muscle – Eagel, Churchill, Livingstone, Edinburgh
17. Proprioceptive Neuro muscular facilitation techniques – Knot M. and Voss, Haroer and Row, New York
18. Child with Spina Bifida – Anderson E.M, and Spain B. Methun, London
19. A manual of neonatal intensive care – Robert N.R.C, Edward Arnold, London
20. Pulmonary rehabilitation: guidelines to success – Hoidkina, Butterworth, Boston,
21. Cardiac rehabilitation – Amundsen L.R, Churchill, Livingstone, London

1.20.16. Specialty papers

6) Master of Physiotherapy in Obstetrics and Gynaecology Sciences

MPT (OG) 104: Clinical, physical & functional diagnosis in in OBG Physiotherapy (OGCPFD)

MPT (OG) 202: OBG physiotherapy (OGPT)

MPT (OG) 203: Recent advances and Evidence Based practice in in OBG Physiotherapy (OGRAEB)

SPECIALITY PAPER -ONE

1.20.16.A. COURSE CODE: MPT (OG)-104

MPT (OG) 104: Clinical, physical & functional diagnosis in in OBG Physiotherapy (OGCPFD)

OGCPFD 1.0.1. Course Outcomes:

On successful completion of this subject it is expected that students will be able to-

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in OBG conditions & interpret clinical tests and special investigations commonly used in the diagnosis of these conditions.
2. Generate a primary diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on OBG Physiotherapy practice.
5. Demonstrate a broad range of technical skill in diagnosing the physiotherapy related OBG conditions.

SECTION- A

OGCPFD 1.1. GENERAL ANATOMY AND PHYSIOLOGY OBG

1. Anatomy of female reproductive system and abdominal wall
2. Contents of the pelvic cavity- Pelvic diaphragm, Pelvic floor muscles, Perineum and external genitalia
3. Pelvic axis, position, obstetric diameters and shape, abnormal bony pelvis
4. Clinical biomechanics and patho-mechanics of spine, female pelvis, posture, movement and gait.
5. Ovulation induction, Ovarian function, clinical aspects of ovulation

6. Premenstrual syndrome
7. Polycystic ovarian syndrome
8. Menstruation cycle and other clinical phenomena such as amenorrhea, dysmenorrhea, hemorrhagia, polymenorrhea, oligomenorrhea and hypothalamic pituitary dysfunction

OGCPFD 1.2. PREGNANCY, LABOR AND PUERPERIUM

1. Preconception health, factors affecting conception
2. Conception
3. Physiological changes during pregnancy.
4. Physiology of labor.
5. Physiological changes and physical problems in puerperium.
6. Injuries of uterine support & pelvic joints during labor, Repair of perineum after delivery.
7. Anatomical & physiological changes during postpartum period.

OGCPFD 1.3. CONTRACEPTION, STERILIZATION AND FERTILITY

1. Inject able and implantable contraception.
2. Intra uterine devices.
3. Abortion and Miscarriage.
4. MTP and sterilization.
5. Fertility, infertility, sub fertility.

OGCPFD 1.4. Role of PT in high-risk pregnancy

1. Abortion, ectopic pregnancy.
2. Heart disease in pregnancy assessment.
3. Diabetes mellitus in pregnancy.
4. UTI in pregnancy.
5. HIV in pregnancy.
6. Trauma in pregnancy.
7. Hypertension in pregnancy.
8. Gastrointestinal disorders in pregnancy.
9. Viral exposure during pregnancy.
10. Vaginal birth after cesarean section.

OGCPFD 1.5. UROGYNAECOLOGY SYSTEM

1. Review of mechanism of continence and voiding difficulties.
2. Review of Sexual dysfunction in Urogynecology.
3. Assessment of Urinary bladder dysfunction.
4. Genital Prolapse, Assessment and diagnosis.
5. Other displacements of uterus, assessment and diagnosis.
6. Overactive bladder syndrome, assessment and diagnosis.

SECTION- B

OGCPFD 2.1. THE AGEING FEMALE

1. Anatomical & physiological & psychological changes of Menopause
2. Assessment and diagnosis of Senile osteoporosis & related complications
3. The climacteric- assessment and diagnosis

OGCPFD 2.2. INVESTIGATIONS IN OBSTETRICS AND GYNECOLOGY WITH INTERPRETATION

1. Pregnancy tests and investigations
2. Imaging techniques in obstetrics and gynecology
3. Urodynamics investigations
4. Investigations in endocrinal disorders in females
5. Instrumentation for assessment of Pelvic floor muscles- Perineometer
6. Outcome measures in OBG Physiotherapy

OGCPFD 2.3. MISCELLANEOUS

1. Antenatal Physiotherapy assessment.
2. Postnatal Physiotherapy assessment.
3. Breast function, disorders and assessment
4. Abdominal incisions & assessment
5. Anthropometric measurements
6. Assessment, clinical tests and diagnosis of movement dysfunction and other musculoskeletal dysfunctions during pregnancy and postpartum period.

1.20.16.B. SPECIALITY PAPER TWO

COURSE CODE: MPT-202

MPT (OG) 202: OBG physiotherapy (OGPT)

OGPT 1.0.1. Course outcomes

On successful completion of this subject it is expected that students will be able to-

1. Develop a management plan, generally including some lifestyle factors, in co- operation with the Clinical Supervisor and consider a prognosis that reflects on the patient's problem.
2. Manage a patient in consultation and co-operation with the clinical supervisor, identifying the presenting problem, developing a basic working diagnosis and selecting a treatment regime that considers the presenting problem with some consideration for ethical, practical and pragmatic concerns.
3. Maintain legal (accurate, clear and legible) patient histories, write basic referral letters and recognize the need of further referral in conference with Clinical Supervisor and peers.
4. Discuss the Common exercise prescriptions and their clinical use, and the sequence of treatment and how to advise different sorts of patients

SECTION- A

OGPT 1.1. **PHYSIOTHERAPY MANAGEMENT OF MENSTRUAL PROBLEMS**

1. Nutrition in adolescence
2. Physiotherapy management of puberty disorders

OGPT 1.2. **PHYSIOTHERAPY MANAGEMENT OF MATERNAL MUSCULOSKELETAL DISORDERS**

1. Neck and upper back strain
2. TMJ Pain
3. Thoracic outlet syndrome, costal rib pain
4. Carpel tunnel syndrome
5. Dequervain's diseases
6. Diastasis Recti abdominis
7. Sacroiliac joint dysfunction (anterior and posterior innominate)
8. Symphysis pubis dysfunction
9. Low back pain, piriformis syndrome, coccyx pain
10. Knee and patella dysfunction
11. Nerve palsies, muscle and tendon injuries.

OGPT 1.3. **PHYSICAL THERAPY MANAGEMENT DURING ANTENATAL PERIOD**

1. Early bird classes (Classes taken in first or second trimester about nutrition, Exercise, fetal development)
2. Methods of relieving pregnancy discomfort
3. Preparation for labour
4. Relaxation techniques and Stress Management during pregnancy
5. Aquanatal exercises during antenatal period
6. Exercise prescription during antenatal period
7. Orthotic management during pregnancy
8. Ergonomics in pregnancy

OGPT 1.4. **PHYSICAL THERAPY MANAGEMENT DURING LABOUR PAIN**

1. Perinatal care- Coping strategies for labour
2. TENS in labour
3. Traditional practices related to pregnancy and postpartum management
4. Positions for delivery, types of delivery
5. Pain management and management of discomforts during labour
6. Maternal positions and state during labour
7. Stress management during labour
8. Relaxation techniques
9. Breathing techniques
10. Massage

OGPT 1.5. **PHYSICAL THERAPY MANAGEMENT DURING POSTPARTUM PERIOD**

1. Exercise prescription during postpartum period
2. Lactation management and breast clinic
3. The postnatal period, postnatal exercises and advise
4. Alternative therapies related to pregnancy and postpartum management
5. Schools of manual therapy and joint mobilization techniques
6. Aquanatal exercises during postnatal period
7. Orthotic management during postpartum

8. Stress management during postpartum period
9. Maternal position and state during postpartum period
10. Ergonomic advice in postpartum period
11. Massage techniques
12. Handling techniques of new born

SECTION- B

OGPT 2.1. **GENERAL GYNAECOLOGICAL INFECTIONS**

1. Physiotherapy management for incontinence
2. Physiotherapy management for genital prolapse
3. Physiotherapy management for endometriosis
4. Physiotherapy management for chronic pelvic pain and dyspareunia
5. Physiotherapy management for pelvic inflammatory disease
6. Physiotherapy management for sexually transmitted diseases

OGPT 2.2. **PHYSIOTHERAPY MANAGEMENT FOR SEXUAL DYSFUNCTION**

1. Sexual desire disorders- Hypoactive sexual desire dysfunction, Sexual Aversion disorders
2. Sexual arousal disorders
3. Sexual pain disorders- Dyspareunia, Vaginismus
4. Female orgasmic disorder

OGPT 2.3. **OPERATIVE PROCEDURES AND PHYSIOTHERAPY MANAGEMENT**

1. Principles of surgery and physiotherapy management of intra operative complications
2. Preoperative and post operative care
3. Hysterectomy and Physiotherapy management
4. Fertility awareness and family planning methods
5. Cancer rehabilitation (Breast and Cervical cancer)

OGPT 2.4. **MISCELLANEOUS**

1. Physiotherapy management for musculoskeletal complications during menopause
2. Nutrition for menopause women
3. The method of infection control for physiotherapist working with women's health
4. Assisted reproduction treatments

1.20.16.C. SPECIALITY PAPER -THREE

COURSE CODE: MPT-203

MPT (OG) 203: Recent advances and Evidence Based practice in in OBG Physiotherapy (OGRAEB)

OGRAEB 1.0.1. Course outcome

On successful completion of this subject it is expected that students will be able to-

1. Understand and apply the information regarding recent advances in OBG Physiotherapy for patient care.
2. Search the evidences available for assessment and management of OBG conditions.
3. Apply the evidences available for the management of various OBG conditions.

SECTION- A

OGRAEB 1.1. Antenatal Pilates and Postnatal Pilates Alternative therapies in OBG conditions

OGRAEB 1.2. Alternate approaches to fitness in antenatal and postpartum period

OGRAEB 1.3. Recent advances in outcome measures used in OBG physical therapy

OGRAEB 1.4. Recent advances in evaluation and treatment of maternal musculoskeletal disorders in obstetrics and gynaecology

OGRAEB 1.5. EBP and Recent advances of electrotherapy in OBG Physiotherapy EBP and Recent advances of exercise therapy in OBG Physiotherapy

SECTION- B

OGRAEB 2.1. EBP and Recent advances of Hydrotherapy in OBG Physiotherapy

OGRAEB 2.2. EBP and Recent advances of Thermotherapy in OBG Physiotherapy

OGRAEB 2.3. EBP and Recent advances of Cryotherapy in OBG Physiotherapy

OGRAEB 2.4. EBP and Recent advances of joint mobilization techniques in OBG Physiotherapy

OGRAEB 2.5. Recent Advances in Pelvic Floor Assessment, Devices/ Instrumentation for pelvic rehabilitation

OGRAEB 2.6. EBP of Nutrition in women from adolescence to menopause

OGRAEB 2.7. EBP and Recent Advances in PT following OBG surgeries

OGRAEB 2.8. EBP and Recent Advances in Breast Disorders from menarche to menopause

OGRAEB 2.9. Recent Advances in evaluation and treatment in musculoskeletal conditions – Puberty, Reproductive, Menopausal women

Recommended Reading for OBG Physiotherapy

1. Gray, Henry. Anatomy of the Human Body,
2. C.Guyton, John E. Hall, Textbook of medical physiology, W.B. Saunder company- Harcourt Brace Jovanovich, Inc.
3. D.K.James et al. High Risk Pregnancy-management options, Saunders-An imprint of Elsevier.
4. Margaret Polden, Jill Mantle, Physiotherapy in obstetric and gynecology, Butterworth-Heinemann, Linacre house, Jordan Hill, Oxford, Ann Thomson, Tidy's Physiotherapy, Varghese publishing House, Bombay.
5. Ruth Sapsford, Joanne Bullock-Saxton, Sue Markwell. Women's Health: A Textbook for Physiotherapists,
6. Scientific basis of human movement –Gowitzke, Williams and Wilkins, Baltimore,
7. Clinical biomechanics of spine – White A, and Panjabi- J, B. Lippincot, Philadelphia
8. Physiotherapy in Obstetrics and Gynaecology- 2nd edition- Jill Mantle, Jeanette Haslam, Sue Bartom. Forwarded by Professor Linda Cardow
9. Physiotherapy in Obstetrics &Gynaecology – Polden& Mantle, Jaypee Brothers, New Delhi,
10. D.C Datta -Textbook of Gynaecology. 1st edition
11. Women's Health- A textbook for Physiothgerapists. R. Sapsford J. Bullock. Saxton. S, Markwell. -(W.B. Saunders)
12. Obstetrics &Gynaecologic care in Physical Therapy - 2nd edition - Rebecca. C. Stephenson, Linda. J. O'contuor
13. Clinical Cases in Obstetreics & Gynaecology - Haresh U. Doshi, published by Arihant publishers
14. Advanced in Obstetrics &Gynaecology (vol 2) - ShaliniRajaram, Sumita Mehta, Niraj Goel (Jaypee brothers.
15. Physiotherapy Care for Women's Health – R. Baranitharan, V. MahalaKshmi (jaypee brothers)
16. Williams O Obstetrics- 22nd edition- F. Gary Cunninghan, Krenneth J Leveno, Steven L Bloom.
17. Women's Health- 5th edition edited by Deborah Waller, Ann McPherso (oxford)
18. Het's Manual of Pelvic floor rehabilitation
19. Het's MMT for assessment of pelvic floor muscles.
20. Steven G Gabbe, Jennifer.R. Niebyl Joe Leigh simpson- Obstetrics Normal & Problem Pregnancies - 5th edition- associate editors : Henry Galon, Laura Guetzl, Mark Landson, Eric.R.M. Jauniau

1.20.17 Specialty papers

7) Master of Physiotherapy in Oncology Sciences

MPT (O) 104: Clinical, physical & functional diagnosis in **oncology** physiotherapy (OCPFD)

MPT (O) 202: **Oncology** physiotherapy (OPT)

MPT (O) 203: Recent advances and Evidence Based practice in **oncology** PHYSIOTHERAPY (ORAEB)

1.20.17.A. SPECIALITY PAPER ONE

COURSE CODE: MPT (O)-104

MPT (O) 104: Clinical, physical & functional diagnosis in oncology physiotherapy (OCPFD)

OCPFD 1.0.1. Course Outcome: On successful completion of this subject it is expected that students will be able to-

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in oncology & interpret clinical tests and special investigations commonly used in the diagnosis of these conditions.
2. Generate a primary diagnosis and a list of differential diagnoses consistent with typical presentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on oncological Physiotherapy practice.
5. Demonstrate a broad range of technical skill in diagnosing the Physiotherapy related oncology conditions.

SECTION- A

OCPFD 1.1. Assessment of clinical signs and symptoms, physical and functional evaluation, differential diagnosis of (bone and soft tissue, breast, gynecological, lung,

OCPFD 1.2. GI, head and neck and pediatric cancers

OCPFD 1.3. clinical analysis of cardiorespiratory fitness, posture, gait, movement and movement dysfunction in cancer patients

OCPFD 1.4. Outcome measures and evaluation in oncological Physiotherapy for cognitive impairment and disability, focal disabilities, global measures of disability, motor impairment, ADL and extended ADL tests, Quality of life, pain, stress and anxiety.

OCPFD 1.5. Diagnostic imaging- types of diagnostic imaging techniques in various types of cancer, clinical interpretation and significance (Chest X-Ray, Barium swallow, Barium enema, USG abdomen, Endoscopy, colonoscopy Mammography and mammogram, MRI, Ultra sound, PET and SPECT, CT scan Gastroscopy, Laparoscopy, Pap smear test, bone scan and other diagnostic imaging, fiber optic endoscopy for diagnosis) histo-pathological, hematological, bacteriological investigations. Nuclear and radio imaging.

OCPFD 1.6. Principles of pathological, hematological, bacteriological investigations related to oncological disorders with interpretation.

SECTION- B

OCPFD 2.1. Influence and relation of physical activity, diet, nutrition, life style, obesity and anthropometric measurement in cancer Neuropsychological tests.

OCPFD 2.2. Evaluation of Cancer Complications like Lymphedema, musculoskeletal, neurological, cardio respiratory. Exercise and cancer related fatigue and its evaluation

OCPFD 2.3. Detailed lymphatic system examination

OCPFD 2.4. Medical intervention (radiation, chemotherapy and surgery) in cancer

1.20.17.B. SPECIALITY PAPER TWO

COURSE CODE: MPT(O)-202

MPT (O) 202: Oncology physiotherapy (OPT)

OPT 1.0.1. Course Outcome: On successful completion of this subject, it is expected that students will be able to-

1. Elicit and interpret clinical signs and symptoms of diseases commonly seen in oncology & interpret clinical tests and special investigations commonly used in the diagnosis of these conditions.
2. Generateaprimarydiagnosisandalistofdifferentialdiagnosesconsistentwithtypicalpresentations.
3. Identify normal & pathological anatomy on diagnostic images.
4. Discuss how the serious and common disorders and the specialized areas of medical practice may impact on oncological Physiotherapy practice.
5. Demonstrate a broad range of technical skill in diagnosing the Physiotherapy related oncology conditions.

SECTION- A

OPT 1.1. Assessment of clinical signs and symptoms, physical and functional evaluation, differential diagnosis of (bone and soft tissue, breast, gynecological, lung,

OPT 1.2. GI, head and neck and pediatric) cancers

OPT 1.3. clinical analysis of cardiorespiratory fitness, posture, gait, movement and movement dysfunction in cancer patients

OPT 1.4. Outcome measures and evaluation in oncological Physiotherapy for cognitive impairment and disability, focal disabilities, global measures of disability, motor impairment, ADL and extended ADL tests, Quality of life, pain, stress and anxiety.

OPT 1.5. Diagnostic imaging- types of diagnostic imaging techniques in various types of cancer, clinical interpretation and significance (Chest X-Ray, Barium swallow, Barium enema, USG abdomen, Endoscopy, colonoscopy Mammography and mammogram, MRI, Ultra sound, PET and SPECT, CT scan Gastroscopy, Laparoscopy, Pap smear test, bone scan and other diagnostic imaging, fiber optic endoscopy for diagnosis) histo-pathological, hematological, bacteriological investigations. Nuclear and radio imaging.

OPT 1.6. Principles of pathological, hematological, bacteriological investigations related to oncological disorders with interpretation.

OPT 1.7. Influence and relation of physical activity, diet, nutrition, life style, obesity and anthropometric measurement in cancer Neuropsychological tests.

OPT 1.8. Evaluation of Cancer Complications like Lymphedema, musculoskeletal, neurological, cardio respiratory. Exercise and cancer related fatigue and its evaluation

OPT 1.9. Detailed lymphatic system examination

OPT 1.10. Medical intervention (radiation, chemotherapy and surgery) in cancer

OPT 1.11. Oncology-Epidemiology, classification, symptomatology, patho- physiology and management of different oncological condition

OPT 1.12. Common pediatric oncology conditions and their assessment, signs and symptoms medical management and Physiotherapy treatment

OPT 1.13. Common pediatric oncology conditions and their assessment, signs and symptoms medical management and Physiotherapy treatment.

SECTION- B

Physiotherapy intervention for

OPT 2.1. Head and neck cancers.

OPT 2.2. Breast cancer

OPT 2.3. Cancers of Reproductive system. Bone tumors.

OPT 2.4. Systemic cancers. CNS Neoplasia. Lung cancer.

OPT 2.5. Metastatic cancers Gastrointestinal cancers.

OPT 2.6. Chemotherapy, radiation therapy and adjunct therapy in cancer patients. Physiotherapy management for neuro-musculoskeletal complications due to cancer treatments

OPT 2.7. Physiotherapy management for various dysfunctions (Bowel and Bladder, Sexual, Neuro-musculoskeletal and Nutritional deficiency) seen in cancer patients.

OPT 2.8. Supportive and Palliative therapy, and pain management in cancer and palliative therapy in cancer patients

OPT 2.9. Rehabilitation act and financial aid for cancer patients

OPT 2.10. Psychosomatic conditions in cancer and their management

OPT 2.11. Physiotherapy management in Intensive care units (ICU)of cancer patients

OPT 2.12. Aids and appliances, adaptive functional devices to improve dysfunction in cancer patients

OPT 2.13. FES, NMES, Biofeedback, Various equipment used in oncology Physiotherapy, Muscle re-education approach, Sensory rehabilitation, Myofascial release technique, Inhibitory and facilitation technique, Functional re-education, skill training, A.D.L training, Tapping in oncological conditions. Balance training

OPT 2.14. Problem based learning for various clinical conditions in oncology Physiotherapy

1.20.17.C. SPECIALITY PAPER THREE

COURSE CODE: MPT(O)-203

MPT (O) 203: Recent advances and Evidence Based practice in ONCOLOGY PHYSIOTHERAPY (ORAEB)

Recent advances and evidence-based practice in oncology Physiotherapy

ORAEB **1.0.1.** **Course Outcome:** On successful completion of this subject it is expected that students will be able to-

1. Understand and apply the information regarding recent advances in neuro Physiotherapy for patient care.
2. Search the evidences available for assessment and management of neurological conditions.
3. Apply the evidences available for the management of various neurological conditions

SECTION- A

ORAEB 1.1. Recent advances in oncological Physiotherapy and Evidences in interventions for oncology related impairments.

ORAEB 1.2. Genetic counselling, Stem cell therapy, Gene therapy, Targeted therapy, Immunotherapy, hormone therapy, thermal ablation, radionics, atomics and Nano medicine

ORAEB 1.3. Recent advances in pain modulation and rehabilitation

ORAEB 1.4. Institutional & community-based rehabilitation and vocational rehabilitation in oncological patients

ORAEB 1.5. Recent advancement in oncology Orthosis – prescription and training. Prosthetic management for mastectomy

ORAEB 1.6. Psychiatry problems in oncological conditions and Physiotherapy (BAT, CBT). Psychological aspects of adaptation during various aspects of disabilities Self-treatment, Exercise precaution, management and exercise prescription for home program, Report writing. Conceptual framework for clinical practice. Requirements for medical opinion or treatment, documentation, prescription, management and advice. Protocol writing

ORAEB 1.7. Recent oncological Physiotherapy technique - Mental imagery technique, virtual reality therapy, Pilate's therapy, Hydrotherapy/ Aqua therapy in oncological patients.

SECTION- B

ORAEB 2.1. Impact of cancer treatment on function and its rehabilitation Psychosocial impact on cancer patient, spouse, family members, society

ORAEB 2.2. History of Evidence Based Practice in Physiotherapy, clinical decision making, importance of evidence-based practice, Evidence about diagnosis, prognosis and therapy. Locating evidences, challenges and barriers in EBP.

ORAEB 2.3. Recent advances in Physiotherapy management of head Neck cancer Recent advances in Physiotherapy management of breast cancer

ORAEB 2.4. Recent advances in Physiotherapy management of Bone tumors

ORAEB 2.5. Recent advances in Physiotherapy management of Lung and respiratory tract Cancer Recent advances in physiotherapy management of systemic cancer

ORAEB 2.6. Sports and physical training in oncological conditions

Recommended books for Oncology Physiotherapy

1. Cancer Rehabilitation: Principles and Practice by Michael Stubblefield & Michael O'Dell 1st Edition
2. Cancer Rehabilitation and Survivorship: Trans disciplinary approaches to Personalized care by Joanne L & Patricia Schmitt 1st Edition
3. Palliative Care & Rehabilitation of Cancer Patients (Cancer Treatment and research) by Charles F. Von Gunten 1st edition
4. Textbook of Palliative Medicine and Supportive Care by Edurdo Bruera 2nd edition
5. ACSM's Guide to Exercise and Cancer survivorship By American College of Sports medicine, Melinda Irvin
6. Fatigue in Cancer: A Multidimensional Approach by Maryl Lynne Winningham, Margaret Barton Burke
7. The Concise Guide to Physiotherapy - Volume 2: Treatment edited by Tim Ainslie.
8. Innovations in Cancer and Palliative Care Education by Lorna Foyle, Janis Hostad.
9. Practical Evidence-based Physiotherapy By Rob Herbert 1st edition
10. Oxford Textbook of Palliative Medicine By Geoffrey Hanks, Nathan I. Cherny, Nicholas A. Christakis, Stein Kaasa 4th Edition
11. Legal Aspects of Physiotherapy By Bridgit Dimond 2nd Edition
12. Rehabilitation and palliation of cancer patients: (Patient care) By Herrmann Delbrück 1st edition
13. Physiotherapy a Psychosocial Approach edited by Sally French 1st Edition

14. Everyone's Guide to Cancer Survivorship: A Road Map for Better Health By Ernest Rosenbaum, Holly Gautier, R.N 1st edition
15. Lymphedema: A Concise Compendium of Theory and Practice By Byung-Boong Lee, John Bergan, Stanley G. Rockson 1st edition
16. Contemporary Issues in Women's Cancers By Suzanne Lockwood 1st Edition
17. Rehabilitation in Cancer Care by Rankin 1st Edition
18. Occupational Therapy In Oncology by Cooper 2nd edition
19. Cancer Rehabilitation: An Introduction for Physiotherapists and Allied Professions by Patricia A. Downie 1st Edition
20. Potential & Possibility Rehabilitation at end of life by Jenny Taylor 1st Edition
21. Cancer Pain Management: A Comprehensive Approach by Karen H. Simpson, Keith Budd
22. Exercise and Cancer Survivorship: Impact on Health Outcomes and Quality of Life edited by John Saxton, Amanda Daley 1st edition
23. Physical Rehabilitation by Osullivan.S.B. & Schmitz.T.J 3rd Edition
24. Physiological Basis of Rehabilitation Medicine by Downey.J.A. & Myers.S.J 2nd Edition
25. Krusens Handbook Of Physical Medicine And Rehabilitation Kottke.F.J. & Lehmann.J.F 4th Edition
26. Clinical Decision Making In Rehabilitation by Basmajian.J.V. & Banerjee.G.N 10th Edition.
27. Rehabilitation Medicine by Delisa.J.A.& Gans.B.M 2nd Edition
28. Physical Medicine and Rehabilitation by Braddom.R.L 1st edition
29. Evidence-Based Rehabilitation; a Guide to Practice by Law.M. 1st edition
30. Assistive Technologies; Principles and Practice by Cook.A.M. & Hussey.S.M. 1st Edition
31. Home Rehabilitation; Guide To Clinical Practice by Anemaet.W.K. & Moffa- Trotter.M 1st Edition
32. Manual Of Physical Medicine And Rehabilitation by Brammer.C.M.;Spires.M 1st edition
33. Essential Physical Medicine And Rehabilitation by Cooper 1st Edition
34. Management In Rehabilitation by Schuch C. P & Sekerak D. K 1st edition
35. American Cancer Society Textbook Of Clinical Oncology By Murphy.G.P.;Lawrence.W 2nd Edition
36. Cancer: Principles And Practice Of Oncology By Devita.V.T; Hellman.S. 7th Ed
37. Clinical Onco5l0o0gy; By Abeloff.M.D; Armitage.J.O. 3rd Ed.
38. Bone Tumours (A Clinico Pathological Study) by Vastrad.M.C. 1st edition

39. Therapeutic Exercise by Caroline Kisner 5th edition
40. Exercise Management: Concepts and Professional Practice by Laurel T. Mackinnon 2nd Edition
41. Advances In Exercise Immunology By Laurel T. Mackinnon 2nd Edition
42. Principles Of Exercises In Physiotherapy 2nd edition
43. Kinesiology Of The Musculoskeltal System : Foundations Of Rehabilitation By Donald A. Neumann 2nd Edition
44. Exercise Therapy: Prevention & Treatment Of Disease by John Gormley, Juliette Hussey 1st edition
45. Physical Examination & Health Assessment by Carolyn Jarvis 5nd Edition
46. Practical Evidence-Based Physiotherapy By Robert Herbert, Gro Jamtvedt 4th edition
47. Principles Of Exercise Therapy by M. Dena Gardiner 6th edition
48. Clinical Decisions In Therapeutic Exercise by Patricia E. Sullivan, Prudence D. Markos 2nd edition
49. Therapeutic Exercise : Treatment Planning For Progression Frances E. Huber, Chris L. Wells 1st edition
50. Textbook Of Therapeutic Exercises By Narayanan 1st edition
51. Exercise Management Concepts And Professional Practice by Laurel T. Mackinnon 1st Edition
52. Clinical Exercise Testing And Prescription 1st Edition
53. Evidence-Based Guide To Therapeutic Physical Agents 1st Edition
54. Therapeutic Exercise Moving Toward Function by Lori Thein Brody, Carrie M.Hall 2nd edition
55. Exercise In Health And Disease 2nd edition
56. Aquatic Rehabilitation by Richard Gene Ruoti, David Michael Morris, Andrew J. Cole 1st Edition
57. ACSM Resou5r0c1es For Clinical Exercise Physiology 1st Edition
58. Advanced Fitness Assessment And Exercise Prescription 3rd Edition
59. ACSMS Resource Manual For Guidelines For Exercise Testing And Prescription 4th Edition
60. ACSMS Guidelines For Exercise Testing And Prescription 6th Edition
61. Exercise Testing And Exercise Prescription For Special Cases by James S. Skinner 2nd Edition
62. Therapeutic Exercise by Basmajian.J.V. & Wolf.S.L 5th Edition.
63. Yogic Exercises: Physiologic And Psychic Processes by Ray.D.S 1st edition
64. Fitness Programming And Physical Disability by Miller.P.D 1st Edition

65. Community Rehabilation Services For People With Disabilities by Karan.O.C. & Greenspan.S 1st edition
66. Essential Readings In Rehabilitation Outcomes Measurement by Dobrzykowski.E.A 1st edition
67. Disability Evaluation by Demeter.S.L. & Andersson.G.B.I 1st edition
68. Safer Lifting For Patient Care by Hollis.M. 3rd edition
69. Disabled Village Children by Werner.D. 1st edition
70. Conditioning With Physical Disabilities by Lockette.K.F. & Keyes.A.M. 1st edition
71. Community Based Rehabilitation Of Persons With Disabilities by Pruthvish.S 1st edition

1.20.18. Specialty papers

8) Master of Physiotherapy in Community Rehabilitation Sciences.

MPT (R) 104: **Physiotherapy in Community Rehabilitation Sciences** (PRC)

MPT (R) 202: **Rehabilitation –Assessment, Evaluation and Assistive Technology** (RAEA)

MPT (R) 203: **Physiotherapy in Clinical Rehabilitation conditions** (PCR)

1.20.18.A. SPECIALITY PAPER ONE

REHABILITATION COURSE CODE: MPT (R)-104

Course Title: MPT (R) 104: **Physiotherapy in Community Rehabilitation Sciences** (PRC)

Section-A

PRC 1.1. Definition, Concept, principles & Scope of Rehabilitation, Community, Healthcare delivery system, Health Administration, Institutional based rehabilitation and community based rehabilitation – its principles and differences, multi-disciplinary approach, role of national institutes, District rehabilitation centre and primary health centre. Physiotherapist as a Master Trainer in CBR & IBR.

PRC 1.2. Epidemiology of dysfunctions & advance skills of physical and functional assessment related to Community. Clinical decision-making skill in management of dysfunction

PRC 1.3. Evidence Based Practice & Recent advances in Community Health. Indian Health statistics

SECTION-B

PRC 2.1. Fitness and health promotion –

i. Principles of fitness for health promotion in community,

ii. Nutrition and Diet.

iii. Stress management through yoga and psycho- somatic approaches.

PRC 2.2. Natural calamity & disaster management – Role of P.T. in disaster management team.

PRC 2.3. I.C.F. [Impairment, Disability, Persons with Disabilities and its implications] Evaluation of Disability & Compensation for Persons with disability Act – 1995 and related Government infrastructure.

PRC 2.4. Physiotherapy Ethics –

i. Code of conduct,

ii. Regulatory Agencies and Legal Issues.

iii. W.H.O.`s policies-about rural Healthcare –

iv. Role of P.T.-Principles of a team work of Medical person/P.T./O.T. audiologist/speech therapist /P.&O./vocational guide in C.B.R. of physically Persons with Disabilities,

PRC 2.5. Public health education methods and appropriate media – Public awareness to the various disabilities, communications, message generation and dissipation.

PRC 2.6. Role of Government & NGOs in CBR, inter-sectoral programs and co- ordination, Implementation of the Act.

PRC 2.7. Rights of persons with disability

Specialty 2

COURSE CODE: MPT (R)-202

Course Title: MPT (R) 202: **Rehabilitation –Assessment, Evaluation and Assistive Technology** (RAEA)

SECTION- A:

RAEA 1.1. Orthotics & Prosthetics: definition, classification, bio mechanical principles; assessment and evaluation, prescription & fabrication

RAEA 1.2. Designing & Training of UL, LL, trunk, neck Orthosis, footwear modifications in various conditions

RAEA 1.3. Designing & Training of UL, LL prosthesis in Amputees.

RAEA 1.4. Indications / Contraindications, psychological aspects of its application.

RAEA 1.5. Use of adaptive devices, design & construction e.g. canes, walkers, wheelchairs.

SECTION- B: Industrial Health

RAEA 2.1. Applied anatomy, physiology and biomechanics related to Industrial health.

RAEA 2.2. Clinical decision-making skill in assessment and management of dysfunction related to Industrial health.

RAEA 2.3. Industrial Physiotherapy- prevention of injuries, physiological restoration, rehabilitation in industrial injuries, work station adaptations/ modifications.

RAEA 2.4. Environmental stress in the industrial area --Accidents due to

1. Physical agents- e.g.-Heat/cold, light, noise, Vibration, U.V. radiation, Ionizing radiation.
2. Chemical agents-Inhalation, local action, ingestion,

3. Mechanical hazards-overuse/fatigue injuries due to ergonomic alteration & evaluation of work place-mechanical stresses as per hierarchy –

   i. Sedentary table work –executives, clerk,

   ii. Inappropriate seating arrangement- vehicle drivers

   iii. Constant standing- watchman- Defence forces, surgeons,

   iv. Over-exertion in labourers - common accidents

4. Psychological hazards- e.g.-executives, monotony & dissatisfaction in job, anxiety of work completion with quality,

   i. Role of P.T. in Industrial setup & Stress management- relaxation modes.

   ii. Physiotherapy role in industry – preventive, promotive, curative, intervention, ergonomic and rehabilitative services.

   iii. Ergonomic considerations and health promotion in the industry

RAEA 2.5. Understanding, and anlysing occupation, job description, job demand analysis, task analysis, Employee fitness, job modification, Employment acts.

RAEA 2.6. Vocational Rehabilitation, evaluation & management.

1.20.18.B SPECIALITY PAPER TWO

COURSE CODE: MPT (R)-203

Course Title: MPT (R) 203: **Physiotherapy in Clinical Rehabilitation conditions** (PCR)

SECTION-A

PCR 1.1. Rehabilitation in musculoskeletal conditions, sport sciences and health promotion

PCR 1.2. Rehabilitation in cardio-pulmonary conditions, and health promotion

PCR 1.3. Rehabilitation in Geriatric conditions

PCR 1.4. Rehabilitation in women's Health

SECTION -B

PCR 2.1. Rehabilitation in neurological conditions, movement & psycho-somatic disorders, pediatric conditions

PCR 2.2. General fitness strategies- body mass composition, assessment, obesity and weight control

1.20.18.C. SPECIALITY PAPER THREE

Skills based outcomes and monitorable indicators for Master of Physiotherapy

1.20.14. Competency Statements

1. Analyse and discuss the biomedical, behavioural and social science bases of Physiotherapy and integrate the bases into Physiotherapy practice.
2. Collects assessment data relevant to the client's needs and Physiotherapy practice.
3. Be able to practice in all types of Healthcare setups independently as well as a team member.
4. Be able to screen, assess, diagnose, treat, prescribe and refer a patient independently.
5. Be able to conduct the patient evaluation and assessment as per condition.
6. Assess, analyse, and plan Physiotherapy management.
7. Apply and evaluate Physiotherapy management.
8. Advise patient on appropriate nutrition, exercises, rest, relaxation and other issues
    i. Demonstrate professional practice.
    ii. Demonstrate autonomous Physiotherapy practice.
    iii. Demonstrate the ability to search and retrieve scientific literature
    iv. Demonstrate an understanding of research methods.
    v. Demonstrate the ability to critically analyse scientific literature
    vi. Prepare Report findings of critical analysis in a scientific format

1.20.15. The Table shows Skill based Learning Outcomes and monitorable indicators:

Table 5.6: Skill based Learning Outcomes, knowledge and monitorable indicators

Sl. No.	Learning outcomes	Knowledge/comprehension	Applications / synthesis /evaluation
1.	Analyse and discuss the biomedical, behavioural and social science bases of Physiotherapy and integrate the bases into Physiotherapy practice	i. Be familiar with normal & abnormal patterns of human development and movement. ii. Understand the anatomical framework of the human body including major systems and aspects of the social, cultural, psychological, environmental, spiritual and belief systems influencing human development. iii. Able to understand the concept of health & its contribution to well- ness.	a. Analyse normal and abnormal patterns of human development and movement. b. Demonstrate understanding of structural and functional anatomy. c. Identify anatomical structure from surface landmarks. d. Describe the normal physiological process and the changes throughout the life span. e. Analyse basic human movement. f. Evaluate the significance of healthy lifestyles for patients/ clients
2	Collects assessment data relevant to the client's needs and Physiotherapy practice.	i. Informs the client of the nature and purpose of assessment as well as any associated significant risk.	a. Perform patient assessment technique which includes to know the condition and to gather information about his/her ailment. b. Monitors the client's health status for significant changes during the course of assessment and takes appropriate actions as required. c. Perform assessment procedure safely and accurately , taking into account client consent, known indications, guidelines, limitations and risk- benefit considerations.

Sl. No.	Learning outcomes	Knowledge/comprehension	Applications / synthesis /evaluation
3.	Be able to conduct the patient evaluation and assessment as per condition.	i. Be familiar with different assessment techniques. ii. Able to examine higher motor functions, cranial nerves, ROM, MMT, Muscle tightness, muscle tone, myotome, sensory evaluation, balance, coordination, hand function, functional outcome measures, Physical fitness, cardio respiratory evaluation, posture &gait. iii. Be familiar with special tests. iv. Basic knowledge on radiological findings & other investigations. v. Demonstrate clinical reasoning with choice of assessment and ex- amination procedures	a. Perform patient assessment technique to know the condition and to gather information about his/ her ailment. b. Safely and accurately examines and re-examines a patient using standardized measures. c. Apply pertinent tests and measurements. d. Interpret all assessment findings to allow for identification of the patient's/client's impairments, activity limitations and participation restrictions. e. Interpret findings and reach a differential diagnosis f. Establish a diagnosis for physiotherapy, identify risks of care, and make appropriate clinical decisions based upon the examination, evaluation and current available evidence.
4	Assess, analyse, and plan Physiotherapy management	i. Identify the principles of assessment, clinical reasoning, problem identification, goal set- ting, treatment planning. ii. Be familiar with different assessment techniques and protocols. iii. Know the protocols used in the department. iv. Justify treatment choices with a sound pathophysiological rationale`	a. Develop rapport to obtain history, current health status and previous functional abilities. b. Interpret the patient's/client's verbal and non-verbal responses. c. Determines the personality traits and analyze how the differences in personality influence approach d. Perform patient assessment technique which includes to know the condition and to gather information about his/her ailment.

Sl. No.	Learning outcomes	Knowledge/comprehension	Applications / synthesis /evaluation
5.	Apply and evaluate Physiotherapy management	i. Know the protocols used in the department. ii. Understand and Prevent/minimise risks and hazards during Physiotherapy interventions iii. Establish equipment is within safety check time frames. iv. Demonstrate knowledge of emergency procedures	a. Demonstrate safe, effective and efficient interventions. b. Evaluate the effectiveness of the Interventions
6	Advise patient on appropriate nutrition, exercises, rest, relaxation other issues	i. Explain the impact of exercise and nutritional status of patient during treatment	a. Assess the patient's status after exercise and proper diet.
7.	Demonstrate professional Practice.	i. Demonstrate attitudes and behavior acceptable to society and the profession ii. Practise in accordance with the Standards of Ethical Conduct iii. Explain the health and safety issues for patients and staff iv. Able to deliver safe, effective and timely Physiotherapy interventions v. Recognizes risk & hazards which can happen during intervention. vi. Ability to reflect and evaluate own practice vii. Modify and adapt professional practice in response to evaluation	a. Demonstrate professional behavior. b. Demonstrate safe Practice Plan and show evidence of Professional development.
8.	Demonstrate autonomous Physiotherapy practice	i. Recognize the critical conditions of patients ii. Be familiar with current literature and evidence based best practice	a. Independently assess and treat patients with single or multiple problems which needs physiotherapeutic intervention. b. Demonstrate an ability to refer to other health professionals when beyond the scope of Physiotherapy

Sl. No.	Learning outcomes	Knowledge/comprehension	Applications / synthesis /evaluation
9.	Demonstrate the ability to search and retrieve scientific literature	i. Define search terms Knowledge on available data search resources ii. Identify relevant sources of Research	a. Develop and modify search strategies appropriately complete searches using relevant and available resources such as electronic data bases. b. Discuss different methods of statistical analysis in relation to different research designs. c. Discuss the possible ethical implications and requirements in health research
10.	Demonstrate an understanding of research methods.	i. Have a basic understanding of the value of different research paradigms to Physiotherapy research. ii. Demonstrate a basic understanding of research processes. iii. Understand the ethics of the research process including plagiarism and consent	a. Describe appropriate research methodologies that may be used to examine a variety of research questions. b. Describe the key elements of research design. c. Describe different methods of data Collection. d. Demonstrate knowledge of basic biomedical statistics
11	Demonstrate the ability to critically analyse scientific literature	i. Identify appropriate criteria to assess quality of different types of literature.	a. Demonstrate an understanding of the process of critical review. b. Demonstrate the use of an appropriate critiquing tool to guide interpretation. c. Critically analyse an appropriate selection of scientific papers

Sl. No.	Learning outcomes	Knowledge/comprehension	Applications / synthesis /evaluation
12	Prepare Report findings of critical analysis in a scientific format	i. Be familiar with different writing format depending on the re search methodology. ii. Be familiar with different referencing styles. iii. Knowledge on presentation methods. iv. Integrate the current literature into physio- therapy practice	a. Use standardized writing format b. Cite references using a recognized scientific method c. Demonstrate an ability to synthesise information from several resources d. Demonstrate the ability to communicate research findings using a variety of presentation methods. e. Critique current Physiotherapy practice with reference to contemporary research literature